



Compute Optics Interface (COI): Energy Efficient Photonic Interconnects for AI Compute Scale-up

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Working Group: Physical Link Layer Energy Efficient Interfaces**TITLE: Compute Optics Interface (COI): Energy Efficient Photonic Interconnects for AI Compute Scale-up**

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ABSTRACT:

The rapid evolution of Artificial Intelligence (AI) is driving larger and highly interconnected compute element clusters along with the need to shuffle huge quantities of data between the compute elements. The next generation of larger AI systems requires cost effective, low latency, dense, and energy efficient optical interconnections between local compute elements with sufficient radix to facilitate these data transfers in these new systems.

Hyperscalers deploying these systems have provided key metrics for the next generation of Compute Optics Interface (COI) links leaving the choice of the best architectures to be discussed at the OIF.

This whitepaper reviews the benefits, considerations, and challenges of a variety of solutions for the Compute Optics Interface (COI).

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Document Revision History

This is the first published revision to oiforum.com/wp-content/uploads/OIF-EEI-COI-WP-01.0.pdf, which includes minor fixes for a reference error and a grammatical error.

1 Introduction

The rapid advancement of artificial intelligence (AI) is driving the development of larger, highly interconnected compute clusters that require massive data movement between processing elements. To support this, next-generation AI systems will demand optical interconnects that are cost-effective, low-latency, high-density, and energy-efficient.

The “Compute Optics Interface” (COI) addresses the need for efficient optical connectivity between local AI compute elements, particularly over longer distances where optics provide clear energy advantages.

We begin by identifying where optical links are most beneficial in AI compute architectures. This is followed by an overview of key link types and their respective strengths. We compare various architectures—including co-packaged/chiplet, on-board, and pluggable designs—as well as interface types such as non-retimed, transmit-retimed, fully retimed, and die-to-die (D2D) solutions. A range of optical modulation techniques is also explored. Finally, we examine the trade-offs between electrical and optical interfaces and assess how well each meets end-user performance and efficiency targets.

1.1 The challenge

AI training relies on large quantities of matrix multiplication. Compute elements are the workhorse of AI compute. These compute elements are designed to accelerate the “multiply and add” operations used in AI matrix computation.

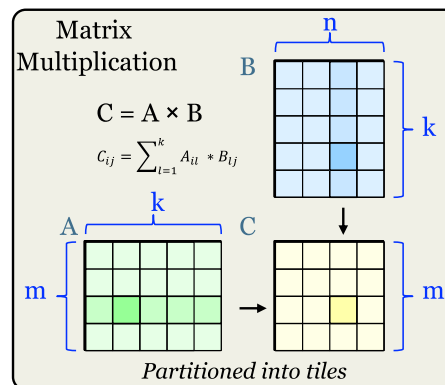


Figure 1-1. AI compute matrix multiplication

Figure 1-1 shows two matrices, A and B , being multiplied to compute matrix C . The matrices are typically too large to be held entirely in the local compute element memory. Therefore, the problems are broken into tiles, and each compute element is assigned a set of tiles to multiply. The result tile in matrix C is not complete until all the different tile multiplications from the various compute elements are complete and summed into tile in matrix C .

The computation is limited by the time to load the values for the input matrices, by the computation speed, by the time to communicate results and weights, and finally by the time to aggregate the overall sums.



The Compute Optics Interface (COI) initiative is driven by a set of pressing challenges in scaling AI systems—challenges that demand new approaches to interconnect design and performance.

A second challenge is the rising energy consumption of AI training clusters. The industry is actively exploring ways to curb this growth—energy-efficient interconnects are a key part of the solution.

However, conventional optical links fall short in several areas. What's needed are next-generation interconnects that deliver higher bandwidth density, lower latency, and greater energy efficiency.

1.2 Types of AI compute links

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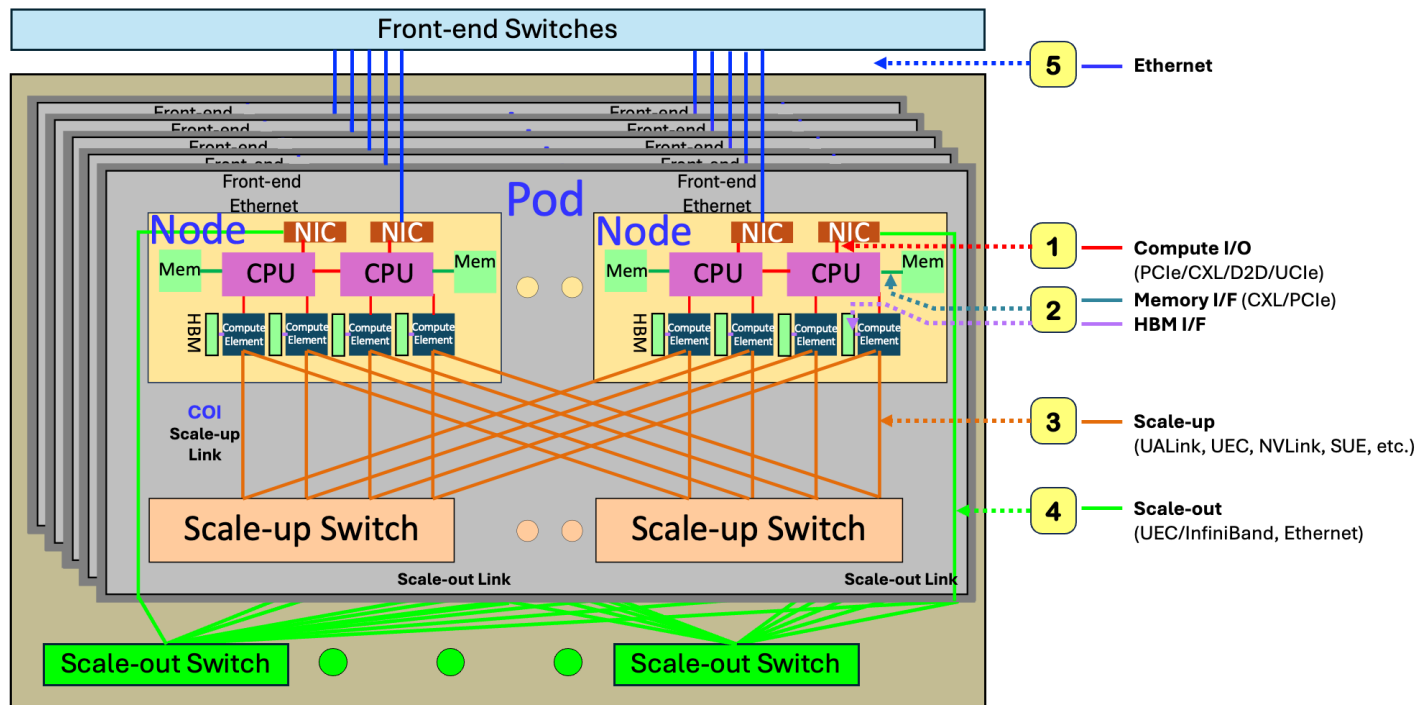


Figure 1-3- AI compute architecture and its interconnects.

Each Pod consists of an array of compute element nodes, as shown in Figure 1-4 employing a variety of different interfaces. Each node typically consists of one or more compute elements connected to one or more CPUs.

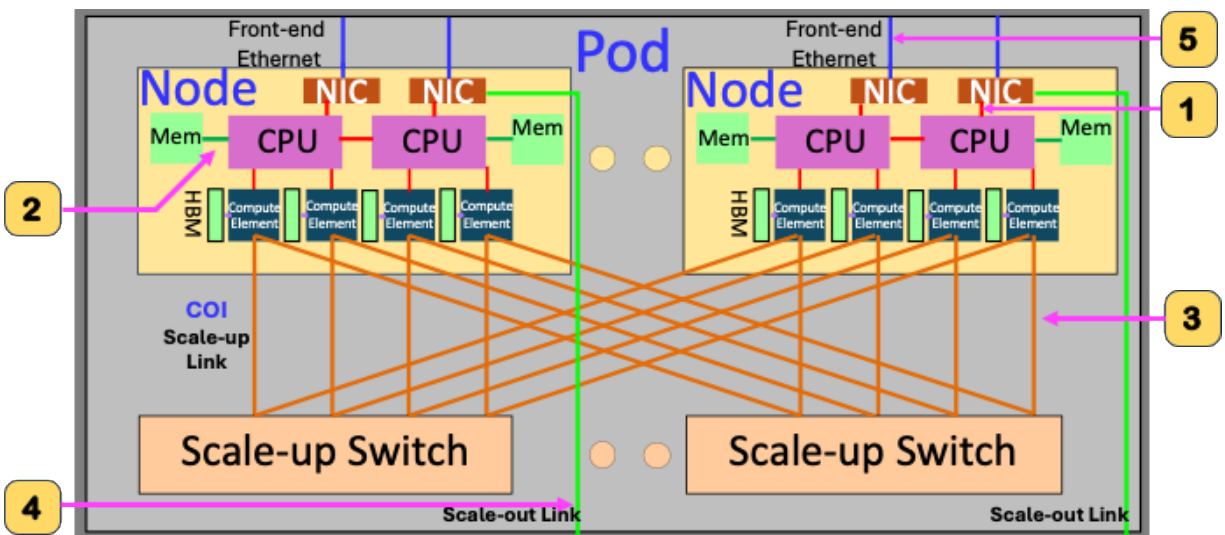


Figure 1-4 - AI compute Pod example

These links are numbered from one to five and color coded. Figure 1-6 shows a description of the various links. The bottom table in Figure 1-6 indicates which link protocols are typically used by each link type.

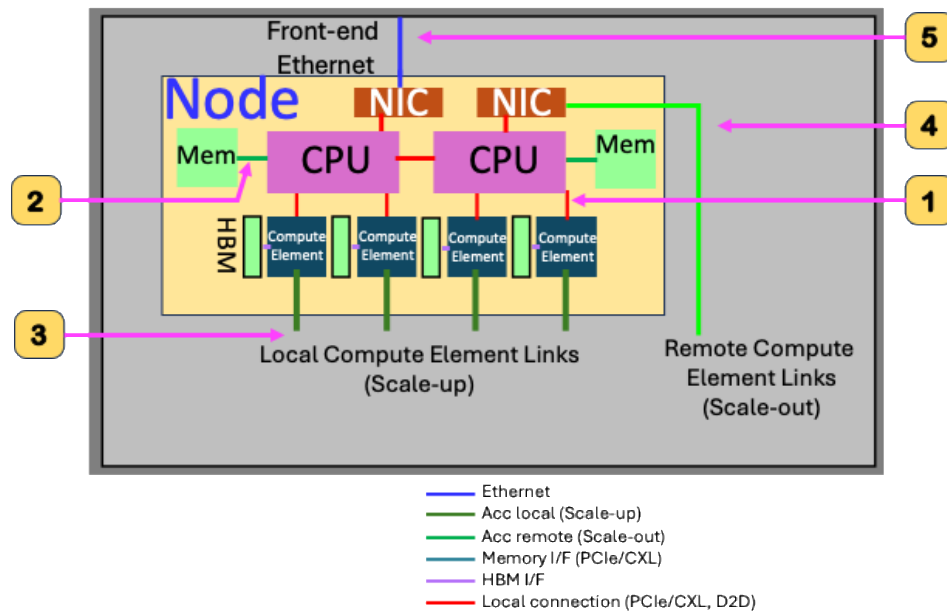


Figure 1-5 - Node Structure

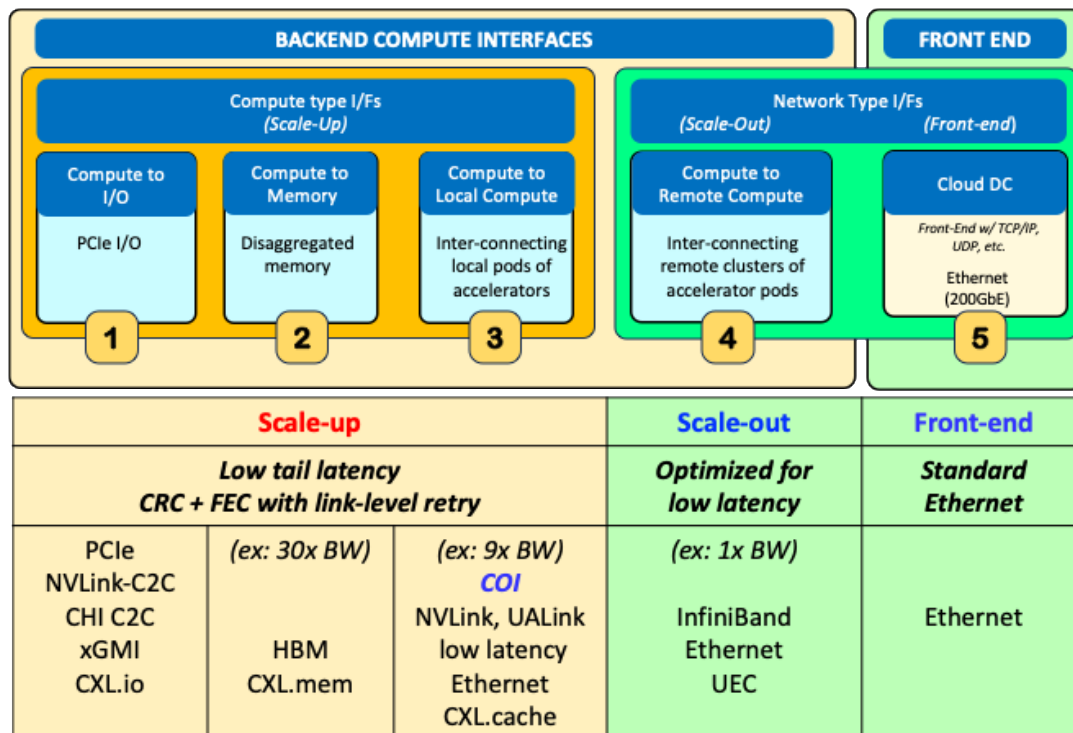


Figure 1-6 - AI compute link descriptions

Links numbered 1 to 5 are used in AI compute clusters. Link type four, the scale-out type links, are typically based on Ethernet or Ethernet variants are used to interconnect between the various local Pods as well connecting these local clusters to remote storage.

The “Compute Optics Interface”, COI, focuses on optical link solutions for link type 3. It enables optical connectivity between compute element nodes supporting I/O coherency across a cluster of local compute element pods. Implementations often incorporate a scale-up switch. These links may transport shared memory access (load/store) requests and responses.

Link types 2 (compute to memory) and 3 (COI) provide various degrees of memory access consistency, for example, avoiding a read before a write has completed, otherwise the results can become corrupted.

Link type 3 (COI) (compute to compute) interfaces usually leverage some kind of flow control which is important to keep the tail latency low. When a corrupt request is detected, the receiver requests a re-transmission. During the retransmission period, no new transfers from that buffer are received, which negatively impacts the overall link transfer rate. Therefore, the COI links benefit from low link error rates.

Figure 1-6 also shows a few relative bandwidths for reference. Note that link type 3 (COI) in some common architectures requires ~9x more bandwidth capacity than that of link type 4 (scale-out).

Figure 1-7 shows several key requirements of the various link types for the different applications summarized from the Optical Internetworking Forum’s (OIF) work [1] on next generation energy efficient links requirements. Note that some boxes may have more than one target requirement. Latency is shown with multiple targets with “latency-A” preferred for latency sensitive applications. Likewise, multiple energy utilization targets are provided with “EU-A” preferred for the lowest power solutions.

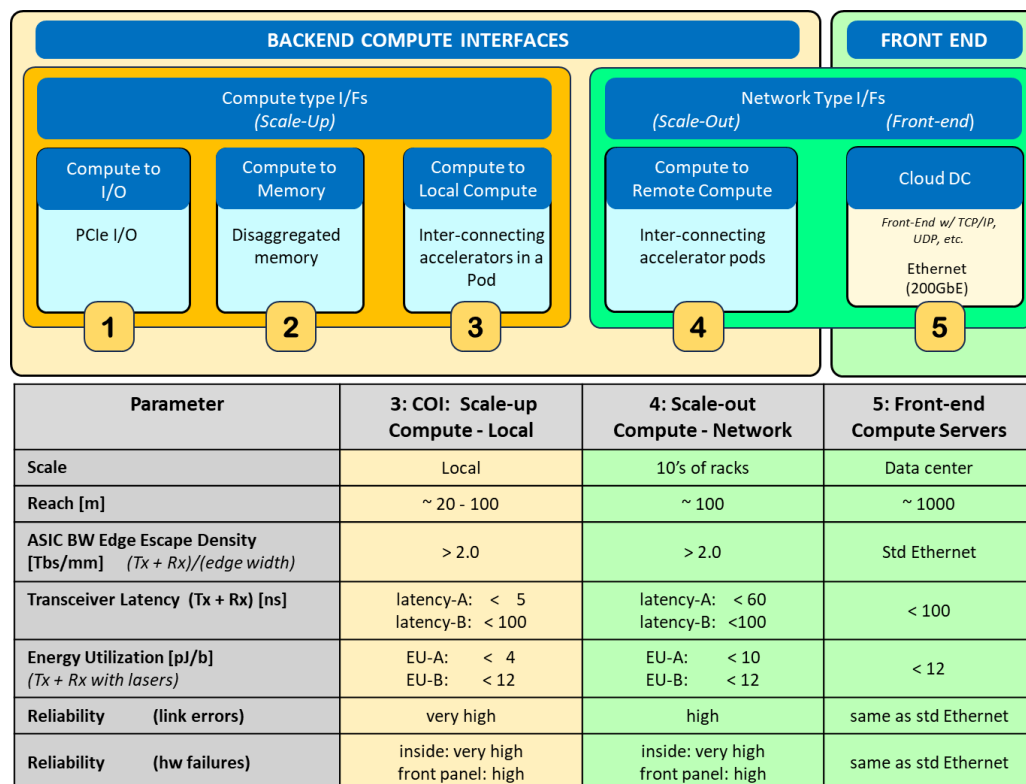


Figure 1-7- AI compute link requirements

In addition, the relative cost of the optical solution should be roughly comparable with copper solution.

1.3 Naively replacing passive copper links with standard pluggable optics

Figure 1-8 shows a representative seventy-two-compute element system in a single rack. In the current generation, the local links are interconnected with copper cables consisting of over 5,000 differential pairs carrying current market leading CEI 224G signals.

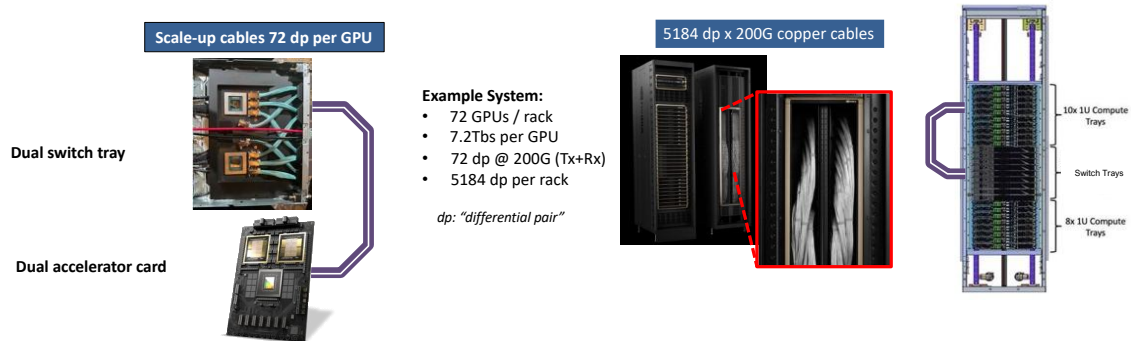


Figure 1-8- Local AI compute example

If one naively replaces the current copper links with CEI-112G-based-800 Gb/s retimed optical modules, the rack would use 1296 optical 800GbE (4x 200G) retimed modules with an additional power consumption approaching 16 kW assuming 12.5 W per module; an astonishing amount of additional power for a rack. In addition, traditional optical modules are less reliable than the passive copper cables, so the system failure rate would increase to ~100,000 FIT assuming ~ 75 FIT per module. Clearly an optimized approach is needed to deploy optics for the local compute elements' interconnects for the next generation of systems.

2 Electrical Interfaces at the ASIC

2.1 Transceiver locations

Figure 2-1 shows three types of optical transceiver placement with two types of electrical interfaces useful for 224G rates and perhaps beyond.

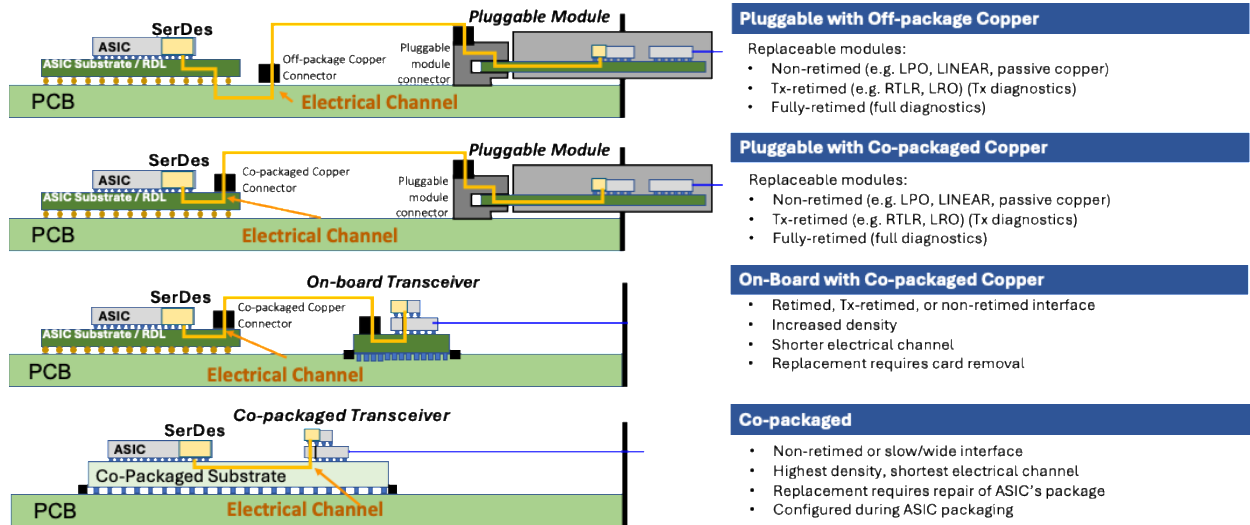


Figure 2-1 - Optical transceiver interconnects

These diagrams employ copper cables for interconnection. The off-package copper solution enables a packaged ASIC to be used with the copper cable. Co-packaged Copper (CPC) requires the ASIC to be co-packaged with the copper connector and avoids routing the high-speed electrical channel through the Host PCB and limits the electrical channel insertion loss for these high-speed data rates.

The top two diagrams show a pluggable transceiver implementation using copper cables. Less often used are on-board-optics as shown in the on-board diagram. Finally, the bottom diagram shows a co-packaged solution. Each of these solutions has differing electrical channel losses between the ASIC and transceiver resulting in differing energy utilizations as well as a variety of other tradeoffs that are discussed later sections.

2.2 Electrical interfaces between the ASIC and the transceiver

Scale-up interfaces may consist of electrical and/or optical links. Figure 2-2 shows a variety of interconnect options utilizing high-speed electrical interfaces.

The top architecture, “Passive Cu Cable”, is what is typically used in current generation interfaces. These electrical links are ideal for short reaches, and passive copper cables are energy efficient but have limited reach (e.g., ~1.5 m for 224 G/L). Table 2-1 shows the historical evolution of long reach SerDes Implementation Agreement capabilities over time.

Table 2-1 – Historical Evolution of the CEI-LR Projects

OIF CEI Projects	CEI-56G-LR	CEI-112G-LR	CEI-224G-LR	CEI-448G-LR
Timeline	2014-2017	2017-2021	2021-	2026-
Modulation	PAM4	PAM4	PAM4	TBD
Insertion Loss	30 dB at 14 GHz (Ball – Ball)	28 dB at 28 GHz (Ball – Ball)	40 dB at 56 GHz (Bump – Bump)	TBD
Reach Objectives	3m copper cable	2m copper cable	1m copper cable	TBD

To build larger Pods, longer-reach connections are required, potentially achieved by adding more retimers (at the cost of increased power) to electrical links or by transitioning to optical links.

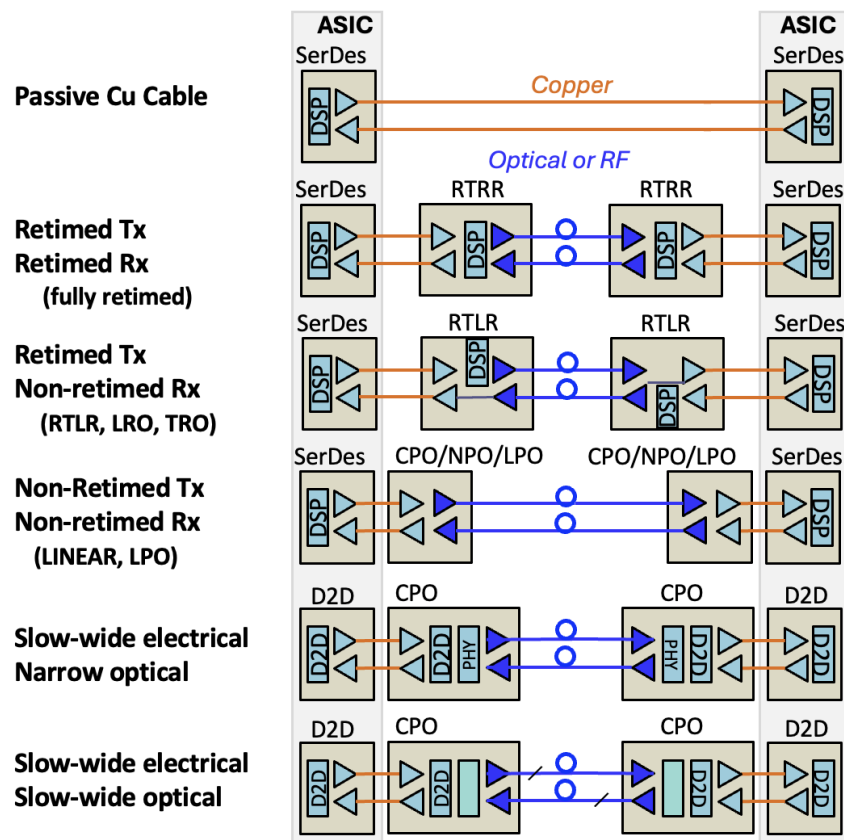


Figure 2-2 - Types of electrical interfaces at the ASIC

The last four architectures in Figure 2-2 illustrate different optical link designs, each varying in their retiming approach. The “Retimed Tx, Retimed Rx” configuration—featuring retimers at both the optical transmitter and receiver—is the most widely deployed. It offers strong diagnostic capabilities and the highest robustness, but also has the lowest energy efficiency among the optical transceiver architectures shown

The “Retimed Tx, Non-retimed Rx” architecture includes a retimer only at the optical transmitter, relying on the electrical SerDes receiver for signal equalization. It supports transmitter-side diagnostics, offers moderate energy efficiency gains, and is expected to deliver reasonable robustness.

The “Non-retimed” architecture eliminates retimers entirely, featuring a linear transmitter and receiver. While it offers limited diagnostic capabilities and is the least robust of the options, it delivers the highest energy efficiency among the architectures.

Lastly, the two “Slow Wide I/F” architectures employ a wide, low-speed electrical interface to the optical transceiver, with the high-speed SerDes integrated within the transceiver itself. This short D2D approach effectively relocates the PHY function to the transceiver in the case of a narrow optical implementation. The optical interface can alternatively support a lower-speed, wider bus of optical lanes.

2.2.1 High-speed serial electrical interfaces

A primary determinant in energy utilization is the degree of additional equalization needed to compensate for the electrical channel loss between the ASIC and the optical transceiver. Figure 2-3 shows the approximate tradeoff between energy utilization of the optical transceiver and the electrical channel insertion loss between the ASIC and the optical transceiver for a typical optical 224 G/L Ethernet link.

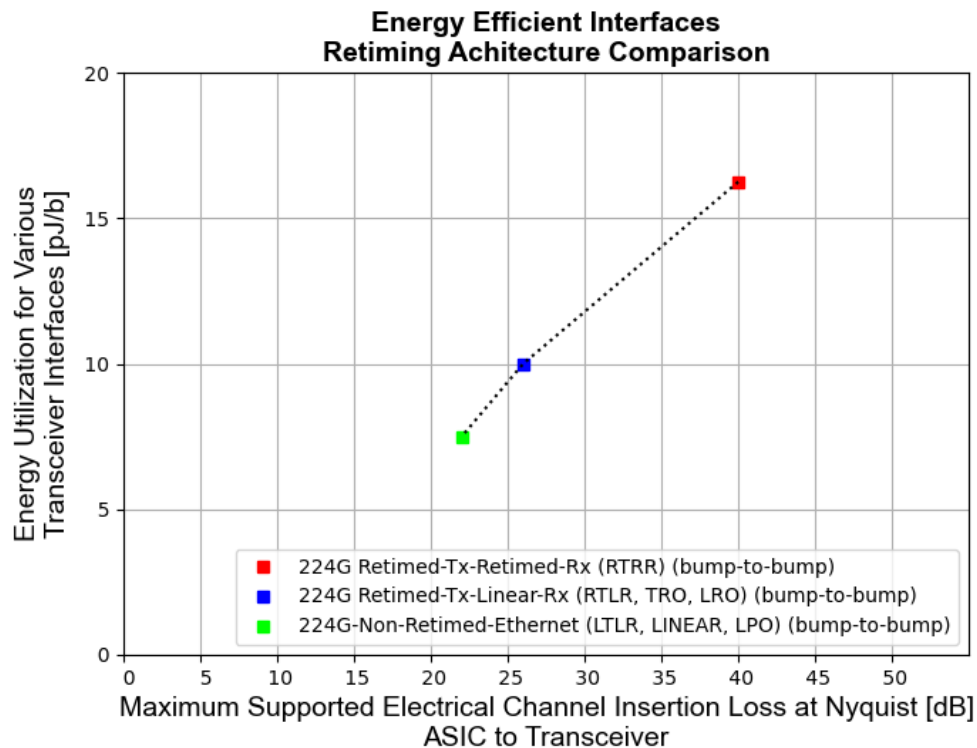


Figure 2-3 – Rough retiming architecture comparison for high-speed serial electrical interfaces

The top diagram in Figure 2-1 (“Pluggable”) can be used with any of the architectures referenced in Figure 2-3. The energy utilization increases as the maximum channel loss increases. For example, an electrical interconnect with a loss of say greater than 25 dB (bump to bump) or more can easily be handled by a

fully retimed (Retimed-Tx-Retimed-Rx) module at the expense of consuming significantly more power in comparison to the lower power alternatives.

With a sufficiently linear optical transceiver and associated well behaved electrical channel (low insertion loss deviation (ILD), crosstalk and reflections), the equalization in the Rx SerDes can compensate for the linear distortions including the optical portions. Transitioning to a lower loss electrical channel enables lower power options. Supporting channels with losses larger than the industry specification is feasible with additional mitigations and with perhaps trading off some degree of interoperability or robustness.

The half retimed (Retimed Tx-Linear-Rx) 224G channel supports up to 26 dB loss (bump-to-bump) as currently envisioned. The retimer function is located only in the near side transmitter (see Figure 2-2) and restores the signal before converting to an optical signal. The far side ASIC's SerDes equalizes the signal between the Tx's retimer and the ASIC.

The degree of linearity is a key consideration. Some optical modulation approaches such as Mach-Zehnder modulators (MZMs) are highly linear whereas others, such as those using direct laser drive current modulation, may be less suitable. In addition, modulators convert the electrical signal voltage or current to an optical power essentially "squaring" any non-linearity present in the modulator's electrical drive signal. This makes the distortion products dependent on the signal level and challenging for a linear equalizer in the SerDes receiver to adapt. Additional equalization techniques such as MLSE/MLSD can be used by the receiving SerDes to mitigate the non-linear distortions.

Fully or partially eliminating retiming in transceivers can reduce optical transceiver power consumption but may also limit diagnostic capabilities that are available in fully retimed implementations. The lack of diagnostics can compromise link robustness. However, shorter electrical channels offer more consistent performance, enabling lower-power solutions with robust signaling margin.

Next generation linear redrivers in transceivers can provide significant gain and reach extension with little or no additional power consumption by providing improved equalization. To achieve higher link margins, lane by lane link training/adaptation can be employed. By utilizing link training protocols such as CMIS-LT, optimum Tx FFE, Rx CTLE, and Rx FFE settings can be automatically computed and programmed.

2.2.2 D2D (slow-wide) interfaces

Another option is to use a Die-to-Die (D2D) also called "slow-wide" or "parallel" electrical interfaces in place of a high-speed serial electrical interface. The industry uses a variety of D2D interfaces such as BoW [2] and UCIe [3] with the latter receiving much attention lately. These D2D interfaces are focused on delivering the greatest shoreline bandwidth density through a balance of speed and lane count to meet bandwidth requirements at low latency and power; typically connecting dies that are flip-chipped onto the same packaging substrate. For example, a standard UCIe package might use 16 to 64 lanes per direction, with lane rates up to 32 Gb/s and reach up to 25 mm. The UCIe 3.0 specification also includes an "advanced package" option, which enables significantly higher bandwidth edge density with rates up to 64Gb/s and supports reaches up to 2 mm. However, this option requires advanced packaging technologies to accommodate the finer pitch interconnects.

Figure 2-4 illustrates the difference between a narrow non-retimed optical interface and a "slow-wide" (parallel) interface. An energy utilization comparison between these two alternatives will depend on the specific implementations employed. Note that the transceiver in the "slow-wide" scenario will consume

more power since the PHY moved from the ASIC to the transceiver with a commensurate reduction in the ASIC's power consumption. D2D interfaces are relatively energy efficient and will contribute to a small increase in the energy utilization of the transceiver (< 1 pJ/bit generally).

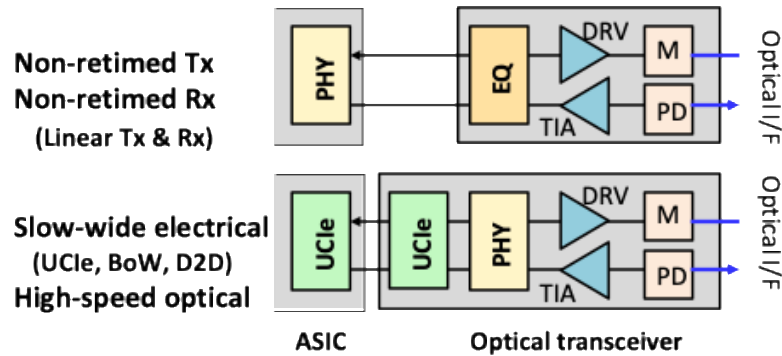


Figure 2-4 - Non-retimed vs a slow-wide architecture

Another important advantage of D2D interfaces is that they support very high edge-bandwidth density on the ASIC as compared to using high-speed SerDes in the ASIC and will be discussed in §2.3.

Since ASIC's are typically large and constrained by reticle limits and power budgets, the compact footprint and lower energy consumption of D2D interfaces help conserve area and power for core ASIC functionality, while offloading the PHY function to the optical transceiver.

2.3 Density metrics

An important requirement (see Figure 1-7) is the ASIC's edge bandwidth escape density. This section illustrates several different approaches and how these approaches impact on the density metric.

The examples in Figure 2-5 and Figure 2-6 shows a width comparison example between the ASIC with 6.4 Tb/s bandwidth and four 1.6T OSFP pluggable modules connected using copper connectors and cables. The 200G SerDes in the ASIC is shown by the white region at the lower edge of the ASIC.

The electrical connection to the modules is made through copper connector and then through copper cables, terminated in this case by 1.6T OSFP modules. The bandwidth edge density achievable at the ASIC edge is limited by the package escape in Figure 2-5 and by the size of the CPC connector in Figure 2-6. Note that ASIC's bandwidth edge density is independent of the size of the pluggable modules for these scenarios.

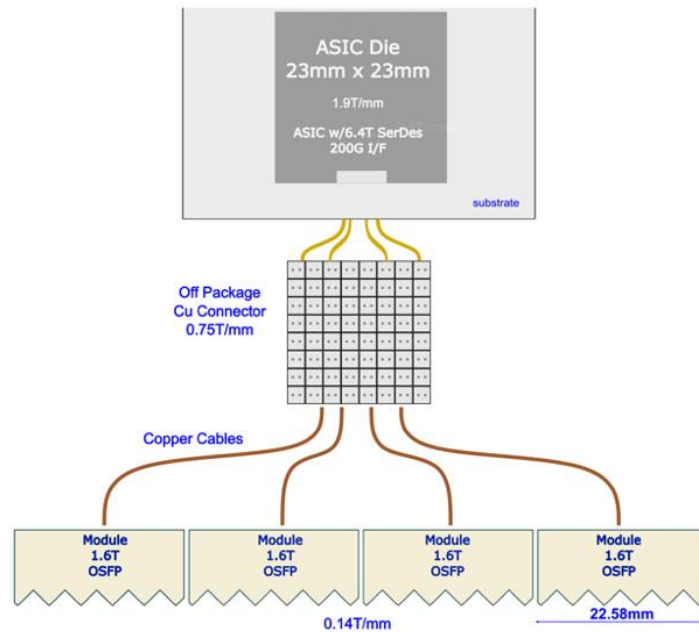


Figure 2-5 - Example of off-packaged copper bandwidth edge density for pluggable modules (<1T/mm).

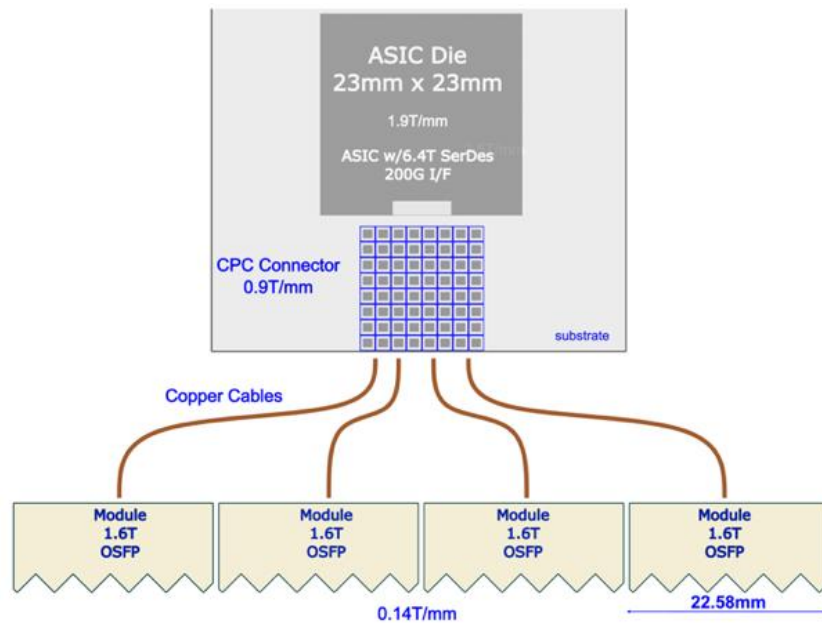


Figure 2-6 – Example of co-packaged copper bandwidth edge density for pluggable modules (~1T/mm).

The example in Figure 2-7 shows several co-packaged scenarios. A hypothetical optical transceiver meeting the minimum bandwidth edge density requirements (see Figure 1-7) is depicted. Three scenarios are shown.

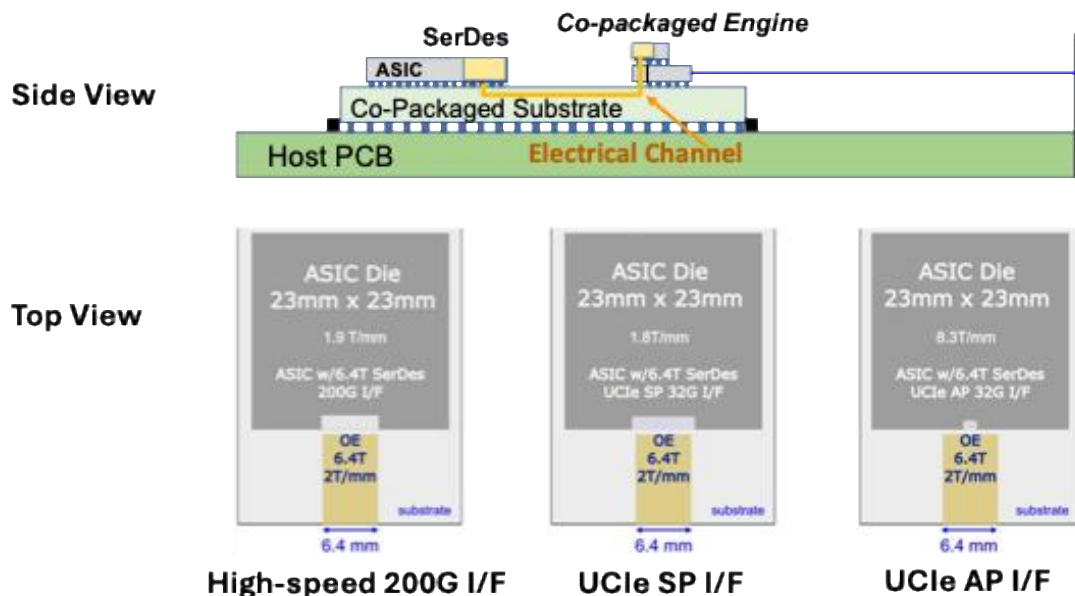


Figure 2-7 – Example: bandwidth edge density comparison for CPO with high-speed electrical (~2T/mm)

First, the “High-speed 200G I/F” (Figure 2-7, on the left) shows the relative sizes of the optical engine compared to the SerDes. In this example, the high-speed SerDes has a 45% higher bandwidth edge density compared to the co-packaged optical transceiver. The drawings in Figure 2-7 show a package substrate for all three configurations although they could use an embedded bridge technology or a silicon interposer technology to support fine pitch attach. The size of the optical engine shown in Figure 2-8 (6.4 mm, derived from the assumed capacity of 6.4T or 32 TX + 32 RX at 200G) is shown as meeting the minimum required edge density of 2 T/mm as set out in the requirements in Figure 1-7.

Next, the “UCle Standard Package Interface” (Figure 2-7, center) uses a D2D interface. The diagram shows the relative size of a UCle 3.0 interface to support 6.4T of bandwidth which roughly matches the width of a 2.0 T/mm co-packaged optical transceiver. Note that the “standard packaged” variant shown uses a ~100 μ m bump pitch commonly used in ASIC packaging and can support up to a 25 mm reach.

The “UCle Advanced Packaged Interface” (Figure 2-7, on the right) variant uses a much denser 25-55 μ m bump pitch (here 40 μ m is shown as an example) and requires the use of silicon interposers or silicon bridges to support the smaller bump pitch. It provides much higher bandwidth edge densities and supports a short 2 mm reach.

Figure 2-8 provides a comparison for the density metric in units of Tbs/mm and for energy utilization in units of picojoule per bit (pJ/b) for various link configurations.

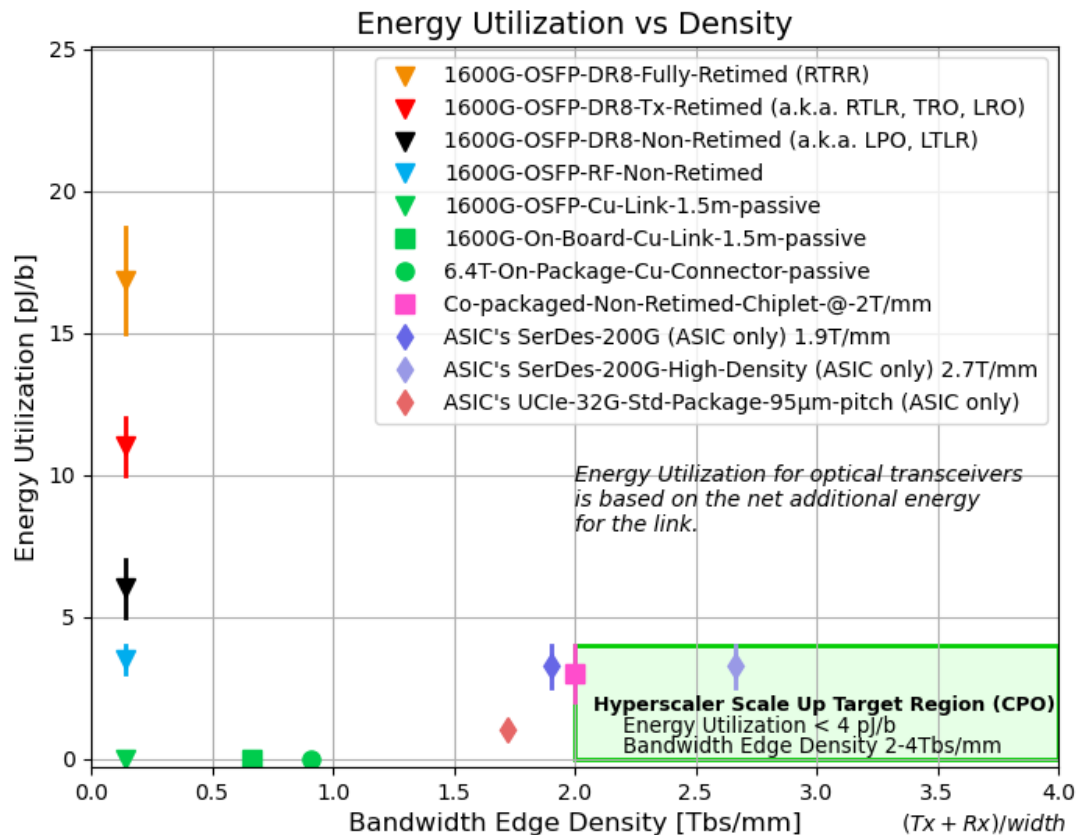


Figure 2-8 - Density metric chart

The leftmost vertical column of triangles represents the current generation of pluggable modules (OSFP type), while the shaded green region indicates the performance target for next-generation modules as defined by hyperscalers. These future modules are expected to use dense copper connections at the ASIC to achieve the required bandwidth edge density. The blue triangle represents an emerging technology using non-retimed linear equalizers combined with RF transmission over plastic cable that can enable reach of 20 m. This provides an intermediate reach capability beyond that of copper and at lower power consumption than that of pluggable optic solutions.

The green square (1600G-On-Board-Cu-Link-1.5m-passive) represents what is currently being used for the local compute links shown by the implementation in Figure 1-8. These passive copper connections have a limited reach of around 1.5 m; thus, longer links are needed to build next generation of modules.

The green dot (6.4T-On-Package-Cu-Connector) provides an example of a co-packaged connector compatible with copper cables.

The pink square (Co-Packaged-Non-Retimed-Chiplet-@-2T/mm) provides an example of an optical transceiver which meets the minimum requirements shown in Figure 1-7.

Finally, the diamonds represent only the ASIC's BW edge density for both a high-speed Ethernet interface (blue diamond) and for a D2D UCle 3.0 interface (standard package) for a 1.6T capacity. The advanced

package UCle variant is not shown as the bandwidth edge density lies far to the right edge of the graph and far exceeds the bandwidth edge density requirements.

2.4 224G Electrical Interface Summary

Table 2-2 - Electrical Interface Summary at 200G with 2.0 T/mm transceiver

Retiming Architecture	Electrical Interface (ASIC to Transceiver)	Optical Transceiver Location (2.0 T/mm)	Achievable ASIC Bandwidth Escape Density [T/mm]	ASIC Bandwidth Escape Density Limiting Factor	Net Interface Energy Utilization [pJ/b]	Max Electrical Channel Reach
Passive Copper	200G	N/A	~ 0.9	Co-packaged Copper Connector	0	~ 1.5 m
Retimed (RTRR)	200G	Pluggable or On-board			15 - 19	> 35 dB
Tx Retimed (RTLr, LRO, TRO)					10 - 12	~ 26 dB
Non-retimed (LTLR, linear, RF)					3 - 6	22 dB
Non-retimed (LTLR, linear)	200G	Co-packaged	~ 2.0	Transceiver	4 - 5	22 dB
Slow-Wide	D2D-UCle standard		~ 1.8	D2D Interface	5 - 6	25 mm
Slow-Wide	D2D-UCle advanced		~ 2.0	Transceiver	4 - 5	2 mm

2.5 Looking ahead to a 448G Common Electrical I/O (Supporting 400G Ethernet)

A key driver for higher-speed electrical interfaces is the need to meet the ASIC's signal escape requirements—specifically, its bandwidth edge density targets. Standards bodies such as the OIF and IEEE are actively exploring the feasibility of 448G electrical links. This work encompasses not only the SerDes design but also the performance of the entire electrical channel, including copper cabling.

An alternative approach is to use a “slow-wide” parallel interface on the ASIC as shown in Figure 2-7 such as UCle which effectively the PHY to the transceiver. The optical link could also be “wide” or “narrow”. In either case, the slow-wide interfaces between the ASIC and the co-packaged transceiver eliminate the need for high-speed copper channels between them.

A key activity will be to compare the 448G electrical-optical link approach with the “slow-wide” electrical interface in terms of the COI requirements. Note that the slow-wide approach provides an option to decouple the optical link from the 448G transmission rate as the optical signal can also be “slow-wide” or high-speed serial.

3 Scale-Up Protocol Choices and the Impact of Electrical Interconnect

3.1 Link protocols

For many generations of server platforms, compute-centric interconnect has focused on PCIe [4] solutions as the primary interface to the CPU offering low latency and memory consistency feature. As high-performance computing (HPC) and artificial intelligence (AI) workloads put more demand on compute element implementations, PCIe continued to remain relevant for intra-chassis links and very short box-to-box links. To meet the scale-up challenges projected, extending PCIe as an optical interface, enables direct memory access (DMA) between devices over extended distances, eliminates the need for complex protocol translations and increases overall system efficiency by employing low-power, and low-latency solutions.

Optical PCIe is compatible with existing compute infrastructure while promoting link extension of up to at least 10 m using a broad range of optical technologies and form factors [5]. PCIe over optics aims to

maintain flexible link architecture with retimed and non-retimed implementations supporting enhanced sideband messaging mechanisms and alignment with industry standard module management features. The PCI-SIG has taken important steps to remain a critical enabler of next-generation scale-up networks, delivering the speed, reliability, and efficiency needed for future workloads.

New recent standardization efforts such as “ESUN” [6], “Ultra Accelerator Link” (UAL) [7], and “Ultra Ethernet” (UEC) [8] are designed on the foundation of the IEEE P802.3dj Ethernet PHYs thereby leveraging Ethernet’s long track record of supporting typical links, including optics.

These new standards differ from the full Ethernet stack in that they employ various improvements to realize low latency with lossless transport and minimize tail latency. UALink and ESUN offer credit-based flow control mechanisms with link-level retry in the data link layer to prevent congestion and minimize tail latency. ESUN optionally offers a priority flow control mechanism which pauses to prevent buffer overflows. All utilize strong forward error correction (FEC) to minimize the number of retransmissions due to transmission errors.

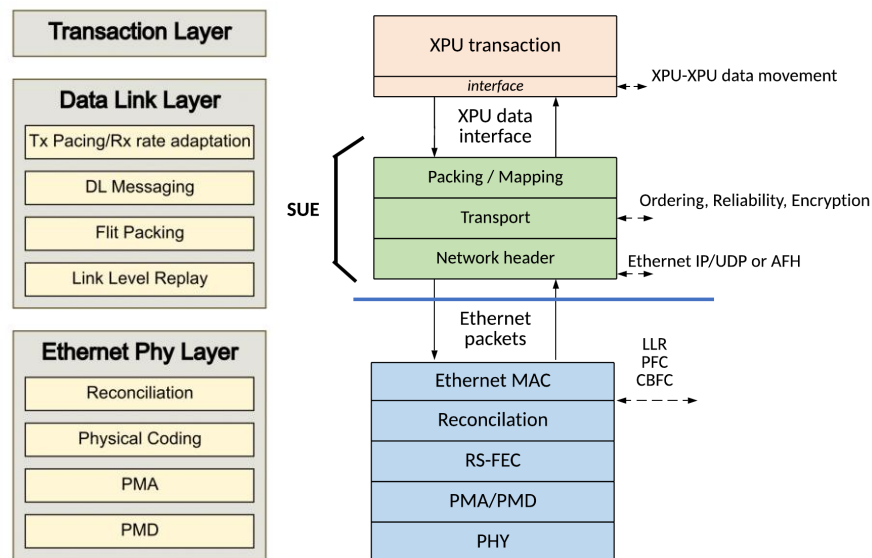


Figure 3-1 – UALink_200 Layer Stack (Left) and SUE Layer Stack (Right, adapted from [9])

Figure 3-1 shows the data link layer stack for UALink_200 as an example of an optimized stack.

Data transfers on these interfaces are subjected to FEC, Ethernet’s RS (544,514) and other FEC schemes supported by different protocols. Some of these protocols follow with a CRC to check for residual errors. If an error is detected, the transfer is not acknowledged and a retry is initiated. Designers are looking at different FEC interleave modes from the standard IEEE 802.3dj specified ones to further reduce latency.

3.2 Latency impact of retiming architectures

Many of the protocols support some form of link-level retry as shown in Figure 3-2. The incoming block or Flit is processed through FEC correction and then followed by a CRC check. If an error is detected during the CRC check, a link-level retry is initiated by replying with a “not-acknowledged” message to the sender. The retry rate will be very small with a sufficiently low bit error rate. However, any retries introduce tail latency as the block or Flit must be eventually retransmitted. The overall retry period will depend on the links’ time-of-flight and on the various blocks in the path.

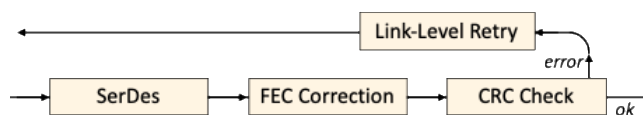


Figure 3-2 - Link-level retry

3.2.1 Example of a UALink_200 Scale-Up link with link-level retry

Figure 3-3 shows a round-trip communication path using an optical interconnect between an compute element and a switch for a UALink_200 link using credit-based flow control. Key blocks that contribute to the link’s latency are shown in blue and includes the time-of-flight (TOF) of the optical fiber based on ~ 5 ns/m.

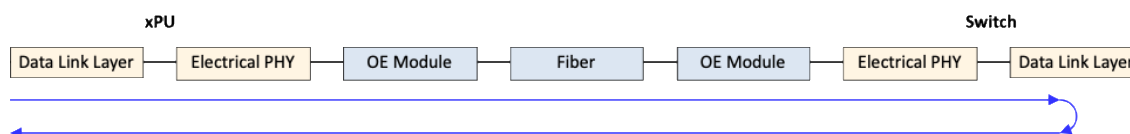


Figure 3-3 – Round-trip through an optical scale-up link with credit-based flow control

The sender’s buffer in the data link layer (DL) is not released until an acknowledgement is received. In the event of an uncorrectable error, the Flit is retransmitted introducing additional delay and thereby negatively impacts the overall link throughput.

The example in Table 3-1 estimates round trip time for a Flit transfer in a link shown in Figure 3-3 and considers the three different retiming architectures discussed previously (non-retimed, Tx retimed, and fully retimed). Note that latencies will vary with implementation with hypothetical values shown in Table 2-2. The estimate assumes optical transceivers at both ends of the link use the same retiming architecture.

Table 3-1 – Example showing impact on round-trip latency for various retiming architectures

Round-trip Latency Estimation Example for an Optical Scale-up Link					
Block	Link Component	Quantity for Round Trip	Non-retimed (LTLR)	Tx retimed (RTLr)	Fully retimed (RTRR)
			CPO	OBO/Pluggable	
1	DL at xPU & Switch, One Direction, Tx + Rx [ns]	1	75		
2	Electrical Ethernet PHY, One Direction, Tx + Rx [ns]	1	40		
3	Optical Transceiver, One Direction, Tx + Rx [ns]	1	5	45	85
4	Optical Fiber (one direction) [5 ns/m]	5	25		
Latency for One Direction (excluding packet length) [ns]			144.8	184.8	224.8
Round Trip (with acknowledgment, excluding packet length) [ns]			289.6	369.6	449.6

Table 3-2 shows the impact on the link throughput due a retry event.

Table 3-2 - Relative Throughput Given One Corrupted Flit to Error-free Case

Relative Throughput of 425G (2x212.5G) Link with one Flit Error Compared to Error Free Case			
Flit Error Ratio	Non-retimed (LTLR)	Tx retimed (RTLr)	Fully retimed (RTRR)
1.00E-02	-18.45%	-22.41%	-25.99%
1.00E-03	-2.21%	-2.81%	-3.39%
1.00E-04	-0.23%	-0.29%	-0.35%

The various retiming architectures impact the link throughput when a Flit is corrupted and a retry is needed.

4 Optical Interfaces Connecting ASICs

4.1 Radix considerations

The radix of a Pod architecture is another important consideration and determines the Pod size. Important design considerations are:

- The feasible I/O bandwidth for the compute element and for the switch IC
- The aggregate link rate which is needed per I/O port on the compute element IC and switch IC

For example, current scale up protocols support several different link capacities with lanes operating at 212.5Gbps.

- A x1 link with 212.5G link bandwidth over 1 lane
- A x2 link with 425G aggregate link bandwidth over 2 lanes
- A x4 link with 850G aggregate link bandwidth over 4 lanes

Full mesh connectivity requires each compute element to connect to all the switches. The number of compute elements in a Pod is therefore determined by the switch I/O bandwidth and the link bandwidth:

$$\text{EQ 4 - 1: } \#Accelerators_{POD} = \frac{Bandwidth_{Switch}}{Bandwidth_{Link}}$$

Allocating a small bandwidth to the link results in slow compute element-to-compute element communication but supports a large number of compute elements in a Pod. Alternatively, allocating a large link bandwidth constrains the number of compute elements that can be interconnected in a Pod.

The hypothetical example shown in Figure 4-1 shows the compute element count for various link rates for various switch I/O bandwidths.

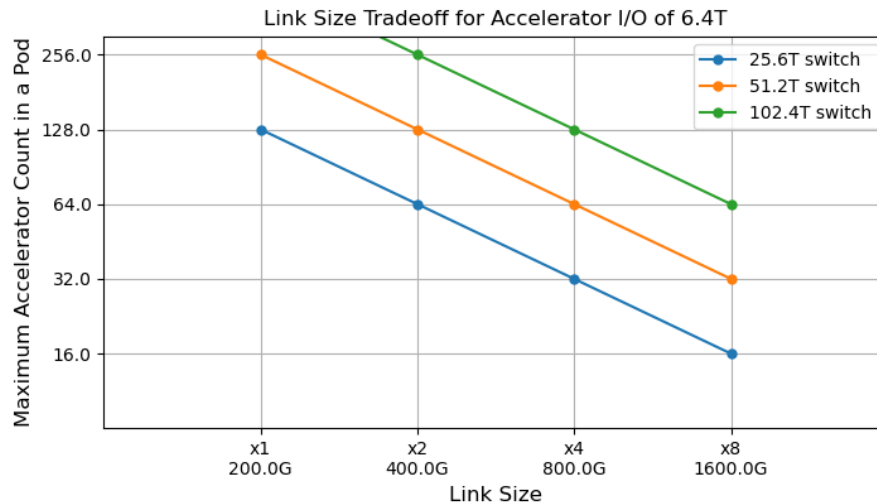


Figure 4-1 – Link size tradeoff for a Pod size

4.2 Optical modulation approaches

The IEEE802.3 optical standards for single mode 200G Ethernet signaling include:

- 1λ per fiber referred to as “DRx” for 500 m reach (O-band)
- 4λ per fiber referred to as “FR4-500” for 500 m reach (O-band with 20-nm grid spacing)

Standard Ethernet transceivers supporting these standards are readily available in pluggable form. However, they may not be optimal for scale-up links due to several factors. For instance, scale-up links typically require much shorter reach compared to the 500-meter reach supported by standard Ethernet optics. To better suit these shorter distances, potential improvements include reducing reach specifications, lowering lane rates, and employing NRZ signaling. Each of these improvements can be realized with multi-wavelength transmission over a single fiber or single wavelength over multiple fibers to create more robust and efficient links.

Notably, compute elements and scale-up switches may benefit from using bookended links specifically optimized for scale-up scenarios.

Some alternative approaches optimized for scale-up have been mentioned:

- Parallel Single Mode (PSM) or Parallel Multi-Mode (PMM), 1λ per fiber
- Coarse Wavelength Division Multiplexed (CWDM) with 2λ or 4λ per fiber (O-band).
- Wavelength Division Multiplexed (WDM) of 4λ , 8λ , or 16λ per fiber with grid spacings in the range of 100 GHz to 800GHz (O-band).
- Bi-Directional transmission (BiDi) with one or more transmit λ s combined with one or more receive λ s on one fiber.
- 2 RF carriers per plastic waveguide

4.3 Hypothetical example configuration of a current generation Pod

Table 4-1 shows an example configuration assuming a full mesh architecture with each link carrying 425G signals. Each link can be composed of multiple lanes. For example, a 425G link might be composed of two 212.5G lanes.

Table 4-1 - Example calculation for a UALink_200 1.0 connected Pod

Parameter	Value	Notes
Accelerator I/O Bandwidth	6.4T	Unidirectional I/O bandwidth for local accelerator connections
Number of Switches	16	Each accelerator connects to all switches in a full mesh
Bandwidth per Link (accelerator BW / number of switches)	400G	Bandwidth per switch coming from an accelerator
Link configuration (link bandwidth / 212.5G per link)	x2	Number of 212.5G lanes comprising the link
Switch I/O Bandwidth	25.6T	Total unidirectional I/O bandwidth supported per switch
Number of Accelerators in the Pod (switch capacity / bandwidth per link)	64	Number of accelerators connected to the switch

UALink_200 is based on copper connectivity at 212.5G. With the introduction of optical links, additional scenarios are available. Some of the possible electrical and fiber connectivity scenarios are shown in Table 4-2 for a 425G link.

The “x2” link follows one of the UALink_200 configurations with two 212.5G lanes grouped as a link. This link could utilize copper-based links such as passive copper cables or an optical links with a variety of configurations.

The “x4” and following scenarios also carry UALink traffic but splits the UALink_200 lanes over a larger number of slower lanes per link. A gearbox in the optical transceiver is needed where the optical lane rate differs from the electrical lane rate. Alternatively, the ASIC can use an D2D interface such as UCle 3.0 as shown in Table 4-2. The D2D interface would connect the ASIC to a co-packaged I/O chiplet (see Figure 2-7). The chiplet is responsible for spreading the traffic over multiple lanes. Slower lane rates can be advantageous as they usually provide better link margin at the expense of using additional lanes.

Table 4-2 Current generation connectivity example

Configurations for 425G per Link	Lanes per Link				
	x2	x4	x8	x16	x32
ASIC Electrical Interface Type	UALink_200 1.0	UCle 3.0	UCle 3.0	UCle 3.0	UCle 3.0
Transceiver Location	Pluggable On-Board Co-packaged	Co-packaged	Co-packaged	Co-packaged	Co-packaged
ASIC Electrical Interface Lane Rate [G]	212.5	32	32	32	32
Optical Lane Rate [G]	212.5	106.25	53.125	26.5625	13.28125
Link Options	Number of Cu Pairs or Fibers/Cores Required (Tx+Rx)				
Cu pair (KR/CR) or 1λ per fiber (PSM/DR)	4	8	16	32	64
1λ per fiber (BiDir2)	2	4	8	16	32
2λ per fiber (CWDM2) and 2RF/waveguide	2	4	8	16	32
2λ per fiber (BiDir4)	1	2	4	8	16
4λ per fiber (CWDM4, WDM4)		2	4	8	16
8λ per fiber (WDM8)			2	4	8
16λ per fiber (WDM16)				2	4

4.4 Hypothetical configuration example for a next generation Pod

One possible scenario for a next generation system using UALink_200 is shown in Table 4-3. It differs from the previous example in that the I/O bandwidths have doubled including the link bandwidth. However, the maximum lane bandwidth for UALink_200 remains at 212.5G so each link has twice as many lanes.

Table 4-3 Hypothetical next generation Pod

Parameter	Value	Notes
Accelerator I/O Bandwidth	12.8T	Unidirectional I/O bandwidth for local accelerator connections
Number of Switches	16	Each accelerator connects to all switches in a full mesh
Bandwidth per Link (accelerator BW / number of switches)	800G	Bandwidth per switch coming from an accelerator
Link configuration (link bandwidth / 212.5G per link)	x4	Number of 212.5G lanes comprising the link
Switch Capacity	51.2T	Total unidirectional I/O bandwidth supported by one switch
Number of Accelerators in the Pod (switch capacity / bandwidth per link)	64	Number of 212.5G lanes comprising the link

The various electrical and fiber connectivity options for this scenario are shown in Table 4-4.

Table 4-4 - Next generation connectivity example

Configurations for 850G per Link	Lanes per Link			
	x4	x8	x16	x32
ASIC Electrical Interface Type	UALink 200 1.0	UCle 3.0	UCle 3.0	UCle 3.0
Transceiver Location	Pluggable On-Board Co-packaged	Co-packaged	Co-packaged	Co-packaged
ASIC Electrical Interface Lane Rate [G]	212.5	32	32	32
Optical Lane Rate [G]	212.5	106.25	53.125	26.5625
Link Options	Number of Cu Pairs or Fibers/Cores Required (Tx+Rx)			
Cu pair (KR/CR) or 1λ per fiber (Parallel MMF/VR/SR and PSM/DR)	8	16	32	64
1λ per fiber (BiDir2)	4	8	16	32
2λ per fiber (CWDM2)	4	8	16	32
2λ per fiber (BiDir4)	2	4	8	16
4λ per fiber (CWDM4, WDM4)	2	4	8	16
8λ per fiber (WDM8)		2	4	8
16λ per fiber (WDM16)			2	4

4.5 Looking ahead to 425G optical

One might expect a future release of UALink that would support 425G lanes as shown in Table 4-5. This scenario is like the scenario in §4.4 (850G links) except that the electrical lane rate is 425G.

The use of 425G serial rates can limit the technology choices available for scale-up links. Whereas indium phosphide (InP) is a widely used material for PAM4 optics at both 100G and 200G data rates and has been demonstrated at 400G serial data rates, silicon photonics IMDD modulators and detectors have -3 dB bandwidths less than that required for 425G with PAM4 signaling. New materials with acceptable bandwidths such as thin-film lithium niobate (TFLN), barium titanate (BaTiO₃), and other technologies are being developed. TFLN based modulators are available but tend to be larger than an equivalent silicon based MZM making it difficult to achieve the co-packaging density target of 2.0 T/mm. These larger or less mature modulator technologies may be better suited for high-density pluggable modules used in scale-up applications, whereas silicon photonics supporting slower-wider optical lanes may be better aligned with scenarios where transceiver size directly impacts the ASIC's bandwidth edge density—such as in co-packaged implementations.

Table 4-5 - Next generation connectivity example with 425G lanes

Configurations for 850G per Link	Lanes per Link				
	x2	x4	x8	x16	x32
ASIC Electrical Interface Type	UALink @ 400G	UALink 200 1.0	UCle 3.0	UCle 3.0	UCle 3.0
Transceiver Location	Pluggable On-Board Co-packaged	Pluggable On-Board Co-packaged	Co-packaged	Co-packaged	Co-packaged
ASIC Electrical Interface Lane Rate [G]	425	212.5	32	32	32
Optical Lane Rate [G]	425	212.5	106.25	53.125	26.5625

Link Options	Number of Cu Pairs or Fibers/Cores Required (Tx+Rx)				
Cu pair (KR/CR) or 1λ per fiber (Parallel MMF/VR/SR and PSM/DR)	4	8	16	32	64
1λ per fiber (BiDir2)	2	4	8	16	32
2λ per fiber (CWDM2)		4	8	16	32
2λ per fiber (BiDir4)		2	4	8	16
4λ per fiber (CWDM4, WDM4)			4	8	16
8λ per fiber (WDM8)				4	8
16λ per fiber (WDM16)					4

4.6 Optical connector density

Depending on the transceiver location, optical connectors may be on the face of a pluggable or on-board module, or it may be on the edge of a co-packaged transceiver. The connector density will depend on the space available for the connector. A connector for a co-packaged transceiver should strive to meet the 2.0 Tbps/mm ASIC bandwidth edge density target while the density for a larger module is much more relaxed as shown in Figure 2-6. Some transceiver implementations may need a single row of fibers while others can support multiple rows. Finally, transceivers utilizing external laser sources will have additional fiber connectivity requirements to bring in the laser power to feed the modulators.

Multi-core fibers, where a fiber could have 4 or more cores, is an interesting technology to reduce fiber footprint. However, as seen earlier in this section, radix is important as each compute element needs to connect to every switch. Other implications of connector density is further discussed in Section 6.1 and 6.3.

4.7 Optical Interface Summary

Optical interfaces for connecting ASICs in AI compute clusters focus on achieving high bandwidth, low latency, and energy efficiency while addressing Pod size and connectivity challenges. Radix considerations determine Pod architecture, balancing link bandwidth and switch I/O capacity to optimize compute element count. Optical interface approaches will be tailored for shorter reaches, with a focus on improved latency, efficiency and density enabling compact, scalable designs for hyperscale AI systems.

5 Packaging CPO chiplets with an ASIC

5.1 Multi-chip module (MCM) technology

MCM technology is used to integrate multiple chips on a single substrate with chips connected to each other using fast and narrow interfaces. Using advanced packaging technologies such as Chip-on-Wafer-on-Substrate (CoWoS) and Fan-Out Chip-on-Substrate (FOCoS) chips can be connected to each other using slow and wide interfaces to achieve higher bandwidth.

As we integrate multiple chips such as ASICs, HBMs, IO dies and CPO in a single module using advance packaging, the module sizes are reaching sizes closer to 9x-10x reticle pushing the substrate sizes to greater than 100x100 mm².

The ASIC and optical engine (OE) chiplets are mounted on a packaging substrate. Current organic substrate designs support sizes up to 150 mm², utilizing either:

- A low CTE organic core (mainstream), or
- A glass core substrate with through-glass vias (TGV), which is still undergoing reliability validation.

5.2 Assembly

Assembly of optical chiplets within the MCM package involves several options:

- Over-mold
- Non-mold
- Co-grinding (to expose the die's optical coupling interface)

OE chiplets typically employ on-PIC optical connectors. While connector weight imbalance is generally not a critical concern, it is recommended placing OE chiplets symmetrically (e.g., two-sided or four-sided around the ASIC) to enhance manufacturability. One critical factor is the precision passive alignment between the detachable optical connector and the OE chiplets to ensure the good optical link budget. To reduce the sensitivity of the connector positioning, expanded beam components, e.g., micro lens array are packaged to chiplets at wafer level or in post wafer process and are also attached to/integrated in the connector side. The other critical factor is the excess force of detachable connectors, which varies with fiber count and must be carefully controlled. The excess force could be either from spring, clamps, or other fine mechanical structures and could be applied to the chiplet in vertical or lateral direction, as On-PIC connectors can be configured with either vertical or lateral plugs. Such excessive force requires a robust mechanical design to prevent damage to the MCM.

OE chiplets must be tested prior to assembly to ensure they are known-good die. Additional challenges arise with photonic integrated circuits (PICs) that incorporate through-silicon vias (TSVs). These dies are typically thin and large, forming fragile 3D stacks that complicate both assembly and testing, which can negatively impact yield.

For electrical interconnects:

- 100–130 μm bump pitch is used for 224G signals and UCle-S substrate assembly, which is currently in production.
- Smaller bump pitches (e.g., 45 μm for UCle-A) are used on interposers in 2.5D or CoWoS configurations for ASIC-HBM integration, also in high-volume production.

5.3 Reliability

Reliability and yield are critical for CPO packaging. Bump redundancy, especially for power and ground connections, can enhance robustness.

Warpage during the long and complex manufacturing cycle must be minimized to prevent interconnect failures. Additionally, OE chiplets with detachable on-PIC optical connectors may introduce unexpected failure modes in electrical interconnects, potentially compromising the entire package. Strengthened packaging solutions are required to mitigate this risk.

As an interim solution, co-packaged copper (CPC) can be used to connect to off-package optical transceivers, enabling longer reach than current scale-up connections. Today, CPC is deployed with passive copper cables, and using CPC with off-package optical transceivers offers a viable path forward as the CPO ecosystem continues to mature.

5.4 Electrical Assembly Yield Considerations

Maximizing MCM assembly yield is essential to reduce overall cost. Conventional packaging with C4 bumps and large bump pitch provides reasonable yield without requiring redundancy. In contrast, advanced packaging technologies are less mature and must incorporate redundancy to enable repair and maintain assembly yield.

5.4.1 Conventional Package – 2D

Figure 5-1 illustrates the conventional packaging approach. C4 bumps connect the ASIC to the package substrate, while SerDes interfaces link to electro-optic components within the Co-Packaged Optics (CPO) chiplet. These signals can either drive the electro-optic components directly or pass through re-timers or drivers for signal conditioning.

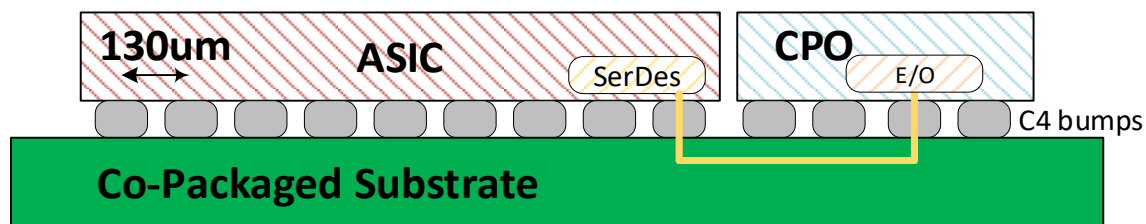


Figure 5-1 - Conventional Packaged CPO

The conventional MCM packaging process is well-established, with assembly-related failure mechanisms thoroughly understood and Design for Manufacturing (DFM) rules in place to optimize yield. One common DFM guideline concerns the placement of C4 bumps at the die corners, as illustrated in Figure 5-2 .

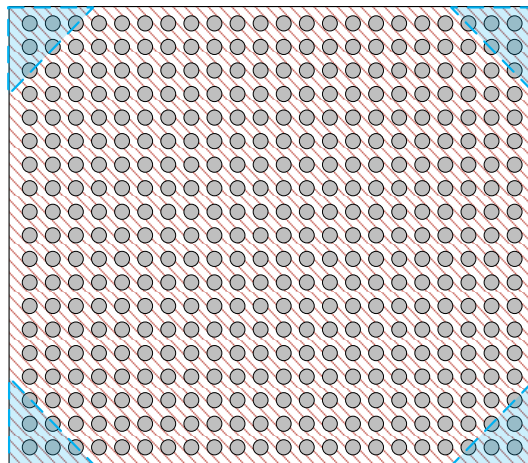


Figure 5-2 - Problematic C4 Bump Areas

Package corners are prone to mechanical stress. To minimize risk, C4 bumps in these areas are typically assigned to power or ground so that a failure does not significantly affect MCM functionality or performance. High-speed signaling C4 bumps should never be placed in these high-risk zones, as defined by DFM guidelines.

5.4.2 Advanced Package – 2.5D

The advanced packaging process uses micro-bumps (μ Bumps) and finer pitch routes to enable 10x more routes per unit length of the shoreline. A standard process has a μ Bump pitch of around 40 μ m and routes with width and spacing of 2 μ m and 2 μ m respectively. There are several variations on how to manufacture and assemble the MCM, but in general the μ Bump pitch and route characteristics are similar between them.

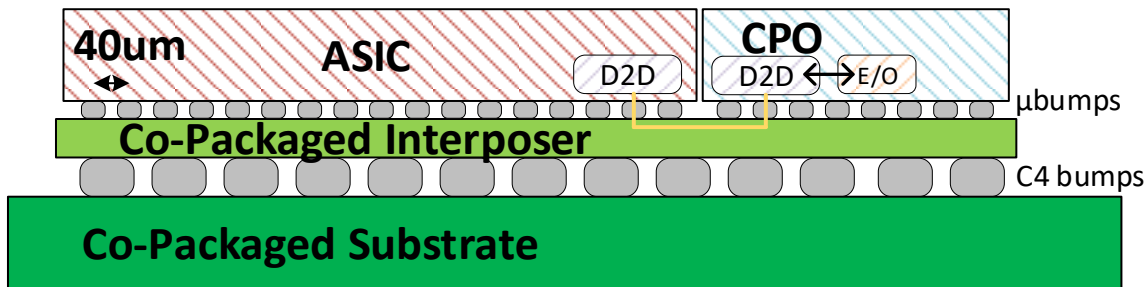


Figure 5-3 - Advanced Package CPO

The primary yield challenge in advanced packages is μ Bump reliability. Failures typically manifest as high-resistance connections or complete opens. These defects occur randomly and are not confined to specific regions of the chiplet, making them harder to predict and mitigate.

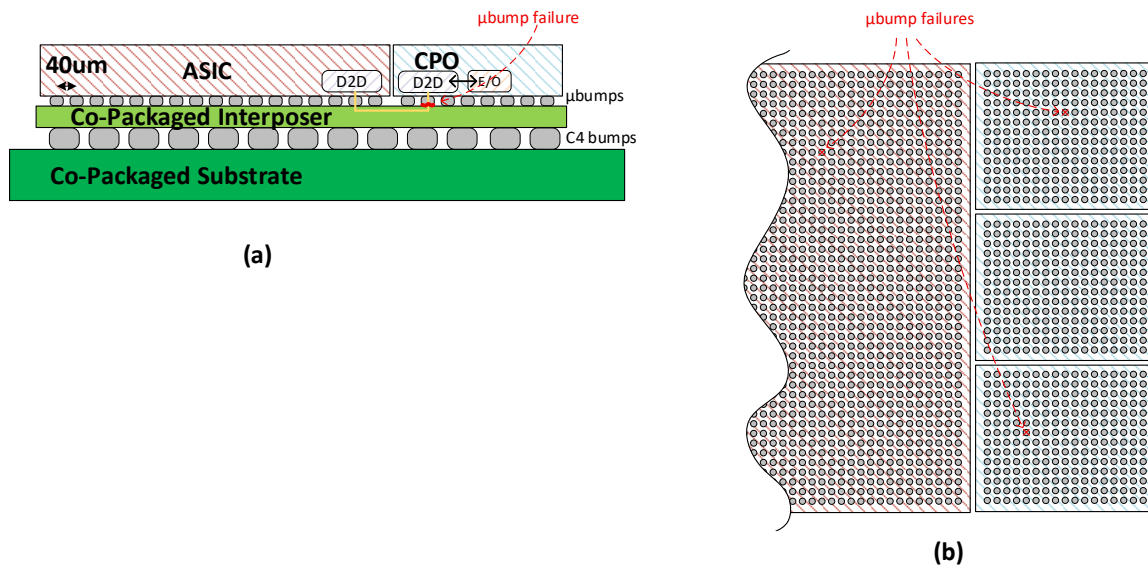


Figure 5-4 - Failure of μ Bumps in Advanced Packages

To achieve reasonable assembly yield, critical signals should use at least two μ Bumps. However, this approach can be costly; for example, in Die-to-Die (D2D) links, it would effectively double the D2D area. Instead, redundancy is applied at the data connection level within D2D IP. Repairs are typically managed within defined bundle sizes, which determine the number of redundant lanes required for acceptable yield:

- **Bundle size ≤ 10 :** One redundant lane is sufficient.
- **Bundle size > 10 :** At least two redundant lanes should be used, enabling repair of bundles up to 50 lanes with high yield (failure rate < 1 in 10,000 parts).

This recommendation assumes a μ Bump failure rate of 1 in 10,000. Figure 5-4 illustrates two commonly used methods for achieving robust assembly in advanced packaging. Failure example (a) shows a dual μ bump approach that can be used for critical signals. This method should be used sparingly because it doubles the μ Bumps (area). Failure example (b) shows a method of adding redundant circuitry for similar structures such as data I/O. This reduces the overhead because the redundant circuitry is shared over multiple I/O. The red “X” marks where the μ Bump fails. The blue arrows show the flow of the signals.

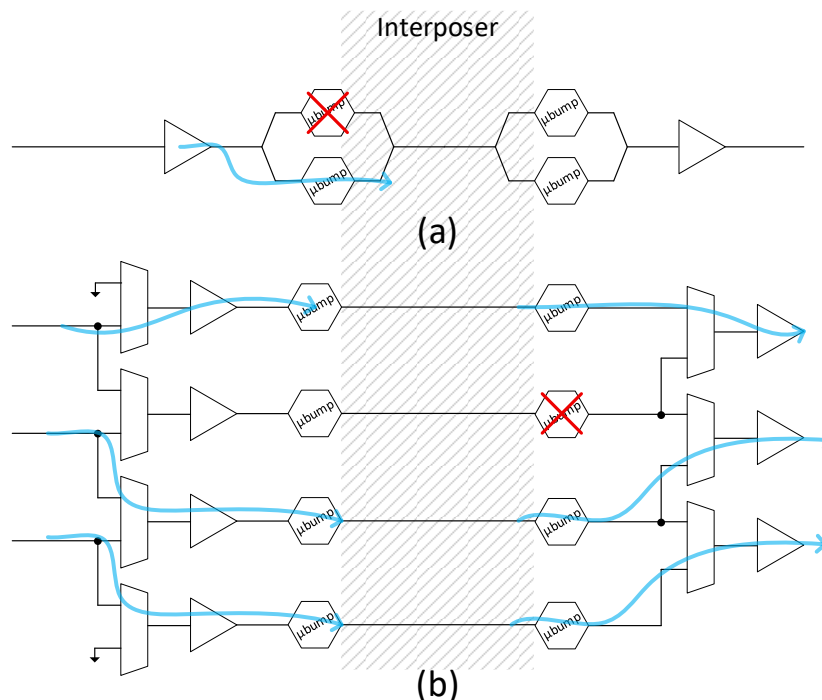


Figure 5-5 – Advanced Packaging Redundancy Methods

Table 5-1 shows packaging relevant properties of standard and advanced packaging interfaces. The UCle 3.0 standard is used as a proxy for the two packaging interfaces.

Table 5-1 - UCle-Standard vs UCle-Advanced Packaging Properties Comparison

UCle Type	Redundancy Needed	Bump Pitch (um)	Data Rate (Gbps)	Linear BW Density (Tx+Rx)/edge width) [T/mm]
Advanced	Yes	40	32	5.2Tbps x 2
Standard	No	130	32	.896 x 2

6 Inter-Rack and Intra-Rack Connectivity

Today's server and rack designs are built around copper cabling infrastructure and as such are governed by the physical and performance considerations of copper cabling. If there is to be a meaningful shift to COI with optical connectivity between local AI compute elements, as this white paper outlines, there needs to be a re-evaluation of server designs, physical backplanes within racks (e.g. optical backplane) and rack-to-rack connectivity to accommodate fiber optic cables.

Copper-centric scale-up networks for AI have been predominantly mesh, torus or a multi-rail single layer of switching, as seen in [10] and [11]. These topologies were chosen because they both supported the ~1 meter reach of copper and map to common collective communication algorithms and parallelism strategies in machine learning. Moving to optics provides new opportunities to enable a variety of network topologies for scale-up systems. This includes higher dimension torus or multi-rail single layer of switching with a larger scale-up domain [12]. The selection of network topology is still an important consideration with optics since different topologies may optimize the link and port definition to target either high bandwidth or high radix. For example, a 4D torus only requires 8 paths per endpoint. As a result, implementations may reduce overall fiber counts through the use of multi-wavelength transmission over a single fiber. Alternatively, a system designer deploying a multi-rail single layer of switching (Figure 6-2) optimizes for high radix and low bandwidth per fiber, since the size of the scale-up domain is determined by the number of ports (i.e. accelerators) that the switch is connected to.

The complexity of optical shuffles within the backplane is similarly dependent on the topology selected. As an example, torus topologies having relatively little need for external shuffles compared to multi-rail single layer of switch, which split out individual fibers across a larger number of endpoints.

6.1 Example configuration of a 512-compute element cluster with fiber connectivity

Figure 6-1 illustrates a scale-up cluster of 512 compute elements connected with a single layer of switching. In this example, a 102.4T switch with a radix of 512 could support 12.8Tb/s of all-to-all compute element connectivity (over fiber) between the 512 compute elements. Each compute element would have 12.8Tb/s of interconnect bandwidth (in each direction) connecting it to each of 64 switches via a 200Gb/s fiber link. From a fiber count perspective, this translates to 64 fiber pairs per compute element and assuming 64 to 128 compute elements per rack, it results in cabling and routing over 4000 to 8000 fiber pairs from each rack.

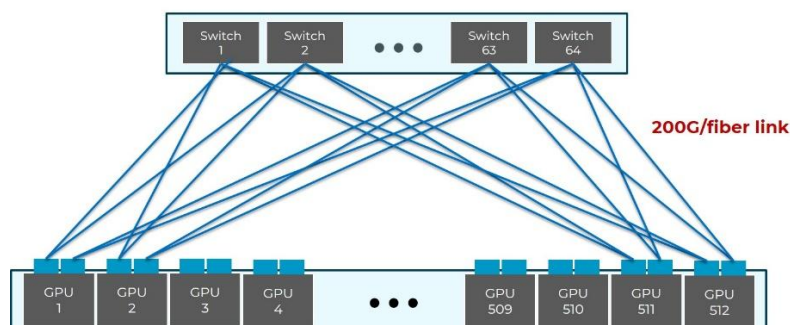


Figure 6-1 All-to-All compute element cluster with breakout of 200G/lane enabled by DR optics and a 512 Radix 102.4T switch.

Note: Bandwidth of each compute element is 12.8T per direction.

Figure 6-2 connected across 8 racks with each rack housing 64 compute elements. One of the key features of moving to optical connectivity is the ability to disaggregate the switches from the server racks that house the compute elements. This in turns enables simplification of rack designs. To achieve this, optical cabling infrastructure is needed to provide the compute element to switch link via rack-to-rack connectivity, as is similar today with scale-out networks. This is illustrated Figure 6-3 for the highlighted section in Figure 6-2. There is fiber connectivity required within the compute element tray to the front or back panel (1). From the compute element tray, there is fiber connection to a fiber shuffle (2). The fiber shuffle enables organization and precise routing of the I/O signals (fibers) from each compute element to the desired switch end-point. Since in this example, for each compute element in rack, an individual 200Gb/s fiber link has to connect to each of the 64 switches that are situated in different racks, it requires organizing I/O fibers from the compute elements within a rack onto cable assemblies, where for a given assembly going to a particular switch rack, each of the 200Gb/s fiber links within originates from distinct compute elements.

Within a switch rack, patch panels (3) enable routing of the incoming cable assemblies from the compute element racks to the appropriate switch-end point via jumper cables (4). Finally, like the connectivity within a compute element tray, fiber connectivity and routing are also required within each switch tray.

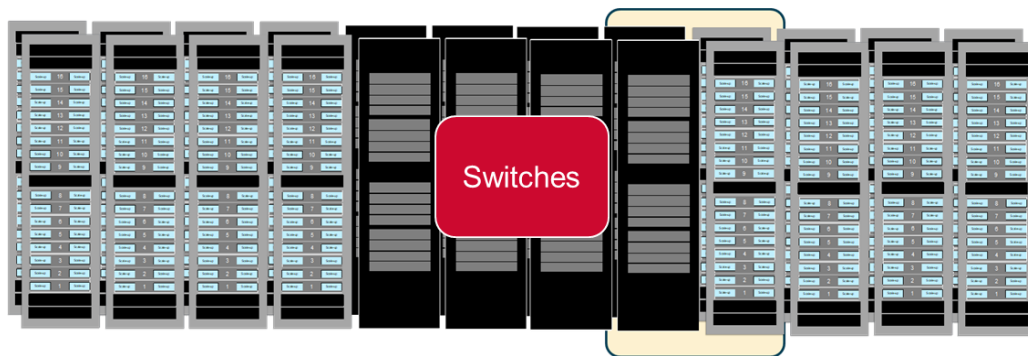


Figure 6-2 - Physical Diagram of 512 All-to-All compute element Cluster with optical connectivity

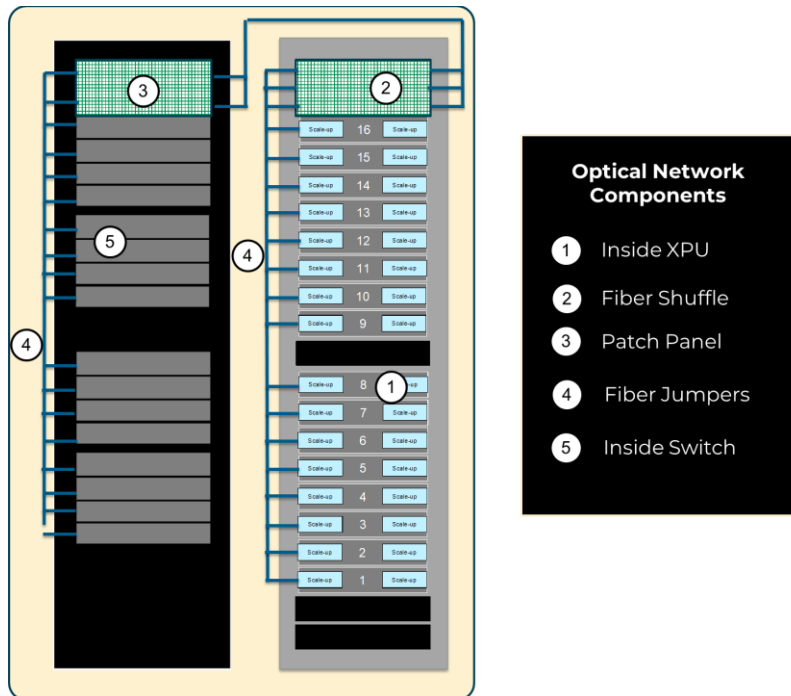


Figure 6-3 - Example connectivity between one compute element rack and switch rack and additional optical network components required

Clearly, at the physical level, this requires infrastructure advancements in connector density, reliability and contamination resistance, structured cabling solutions, shuffle boxes and cable tray designs to implement scale-up architectures with optics. When it comes to rack level connectivity, the key attributes to be investigated for any solution are (1) reliability (2) link budget and (3) Ease of deployment and serviceability.

As mentioned earlier, each server rack housing 64-128 compute elements could have anywhere between 4000-8000 fiber link pairs depending on the total I/O BW per compute element and BW per link. It is not practical to have these connections be placed on the front panel of the servers. Instead, the optical connections should be routed via the backplane. Furthermore, given the large number of fiber connections to each server tray, blind mate optical connectivity is a must. The blind mate technology should also ensure that the mating interfaces are insensitive to dust, which can further improve the reliability of the connections.

It should also be noted that some space on the front panel needs to be reserved for the insertion of plug-gable (replaceable) external laser modules that service the optical engines within the server and switch trays.

6.2 Leveraging Bi-Directional Optics

Another area that could benefit from standardization is the infrastructure surrounding bi-directional (BiDi) optics. For scale-up connectivity, BiDi optics provides a path to maximize fiber efficiency and shoreline density from compute packages. For practical implementations at scale, this will require a framework around the connectivity from the compute element to switch to compute element connection to ensure that the correct wavelength group on the Tx port on one end of the link connects to the correct Rx port on the other end and vice versa. At a physical level as well, there needs to be clear standards on labeling to ensure speed and accuracy in deployment.

Using the example of a 12.8T Optical Engine (OE), the block diagrams in Figure 6-4 and Figure 6-5 illustrate the differences in fiber I/O between a DR and BiDi implementation of a 12.8T optical engine. In the case of DR, since the transmit and receive signals are on separate fibers, a total of 128 fibers (64 Tx + 64 Rx) are required to connect the 12.8T engine to the network. The same connectivity can be achieved by 64 fibers in the equivalent BiDi implementation on account of each fiber carrying 2 wavelengths (e.g. $\lambda_1=1270\text{nm}$; $\lambda_2=1310\text{nm}$). Also, as shown Figure 6-4, there are two equal sets of fibers escaping the engine for BiDi –Group A with transmit on λ_1 / receive on λ_2 and Group B with transmit on λ_2 / receive on λ_1 .

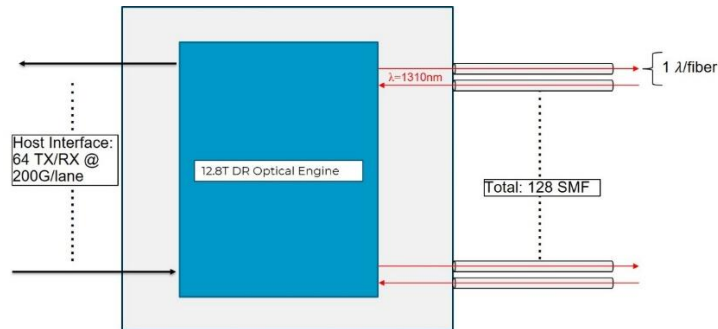


Figure 6-4 - 12.8T optical engine fiber I/O with DR optics

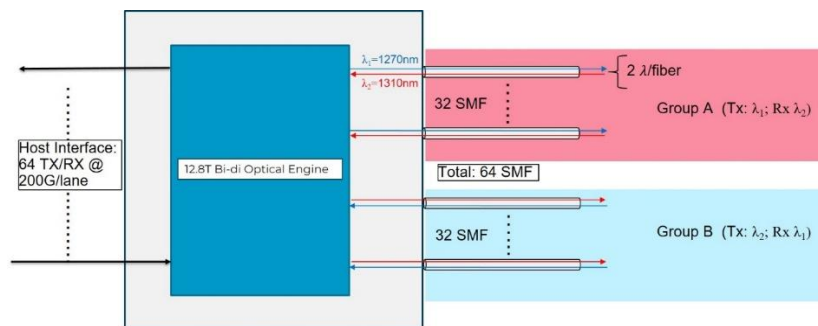


Figure 6-5 - 12.8T optical engine fiber I/O with BiDi optics

Each fiber port from group A must be connected to a Group B fiber port on the other side of the link. This is illustrated in Figure 6-5 where the interconnection between Group A and Group B fiber ports for 2 pairs of OE using BiDi is shown.

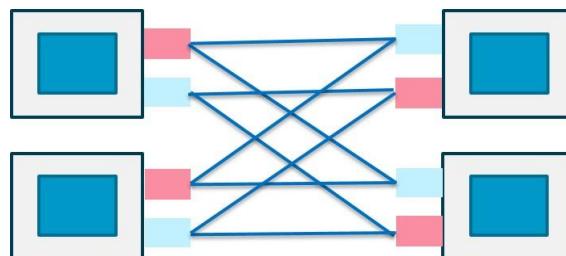


Figure 6-6 - Point to multipoint connectivity using BiDi

Figure 6-7 shows a scale-up cluster with BiDi optics identical in size to the one shown in Figure 6-1. Each switch and compute element have half of its ports allocated for Group A and the other half for Group B. While the basic connectivity including the fiber shuffler as illustrated in Figure 6-3 stays the same, the end

connections need to be correctly matched such that a Group A fiber port on one side of the link matches with a Group B port on the other side.

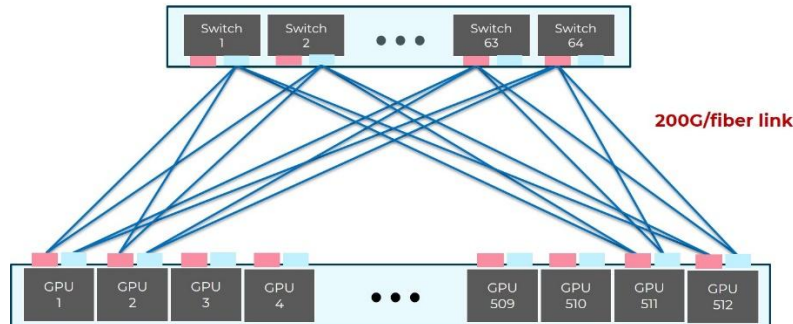


Figure 6-7 - 512 All-to-All compute element Cluster with breakout of 200G/lane enabled by BiDi optics and a 512 Radix 102.4T switch

By building standards around some or all these aspects of rack-to-rack connectivity, the transition to COI can be done at scale within the framework of a multi-vendor ecosystem.

6.3 Optical Connector Density at the Rack

The optical cabling infrastructure for today's back end scale-out networks is already migrating to smaller, denser optical connectors. This accommodates the increased port counts required in rack-to-rack connectivity as compared to traditional compute. These VSFF (very small form factor) multi-fiber optical connectors are 3x denser than traditional MPO connectors and consume minimal rack space, maximizing space for compute elements and network switching. With optical scale-up networks, the fiber density requirements are even higher than in scale-out networks, and VSFF connectivity will be needed in this environment as well.

VSFF connectors support both physical contact and expanded beam technologies. Physical contact connector technology is the predominant optical connector technology used in data center networks, and provides the best performance (i.e. lowest insertion loss). Expanded beam is a technology that expands the optical beam to a larger spot size of light at the connector mating, making it less sensitive to debris, and thus reducing time spent cleaning during deployment and service. As referenced above, due to the extreme fiber counts and backplane location for the connectivity of scale-up compute and switch trays, expanded beam technology may be beneficial in this space. Additionally, some use cases may benefit from a combination of physical contact and expanded beam connectors in the same link.

In addition to traditional front panel and structured cabling solutions, VSFF connector variants are emerging to enable high density connectivity on the card (inside the compute element and switch) as well as for the backplane using optical blind mate solutions. This platform approach enables the industry ecosystem to more easily scale to future scale-up demand.

7 Reliability

As AI compute systems evolve toward dense, memory-coherent architectures, the reliability of interconnects becomes a mission-critical design requirement. Unlike traditional networking environments where packet loss can be mitigated, AI clusters are often tightly synchronized and sensitive to even transient failures. Reliability must therefore be assessed holistically and spanning component durability, system-level architecture, diagnostics, redundancy, and serviceability.

Today these local links are primarily passive copper with little or no redundancy. Optics brings new challenges as the reliability of optics modules is typically worse than a passive cable as discussed in §1.3. In addition, energy efficient solutions may not have all desired diagnostic features to limit downtime during repair.

There are broadly two classes of reliability to consider:

- Link reliability in which at data transfer across the link is corrupted and requires retransmission
- Hardware reliability in which a link or lane can no longer be used to transfer data until repaired

Each of these will be discussed in the following sections.

7.1 Link reliability

Compute elements utilizing memory semantics (load/store operations) ensure memory consistency by ordering I/O operations.

For example, a write must be completed before a read is allowed in the same location.

When the data link layer receives a data transfer, the FEC corrects any errors present. A CRC is performed on the FEC corrected data. If there are no errors detected, the data link layer sends an acknowledgment back to the data link partner who is then free to send the next transfer. However, if an uncorrectable error is detected, the data link layer requests a retransmission (replies with a “not acknowledged”) initiating a retry event. During a retry event, the transmitter resends the data and waits for an acknowledgement before continuing.

These bandwidth impacting situations are often classified as uncorrectable random bit errors or burst errors. In addition, link flaps in which the link rapidly goes up and down also impact throughput. System implementors will likely specify link reliability targets such as frame or FLIT loss ratio (FLR) or a maximum acceptable level of bandwidth reduction. These targets will depend on the system architecture and performance targets. For example, a PCIe 6.0 link might target less than 1% overall bandwidth reduction due to retries.

These errors reduce the effective link bandwidth due to retransmission as described in §3.2. Due to the synchronous nature of local compute, an effective link bandwidth reduction impacts the computation rate of the entire local Pod. Given the significant investment in the AI infrastructure, it is important to employ links with excellent link reliability, and error recovery mechanisms.

7.2 Hardware reliability

Hardware failures have many causes and include underperforming links (high error rate, lowered throughput) or failed links. This class of failures will require hardware replacement.

Each element within the optical interconnect path, ranging from the photonics integrated circuit to light source has distinct reliability concerns and must meet specific qualification standards. Silicon photonics

solutions, for example, could be susceptible to waveguide degradation and electro-optic drift, and are typically evaluated under Telcordia GR-468 and JEDEC protocols. Optical chiplets face thermal and mechanical stressors such as bonding failures and connector fatigue, often requiring automotive-grade standards when applicable. Fiber connectors are prone to contamination, insertion loss, and physical wear, governed by IEC standards while light sources must address mode partition noise and aging per MIL-STD-883 and GR-1221. Sockets and interfaces introduce risks like vibration and alignment errors, guided by IPC or JEDEC standards.

Special emphasis is required on temperature and humidity profiles, particularly in high-power, high-density AI environments discussed here where elevated thermal stress and environmental exposure can significantly accelerate aging and failure rates. Robust component-level qualification, environmental validation, and system-level stress modeling are essential to ensuring the long-term reliability of next-generation compute energy efficient interfaces.

7.2.1 Hardware reliability regimes

Hardware failures are broken down into three regimes: infant failure, random failure, and wear-out failures as shown in Figure 7-1.

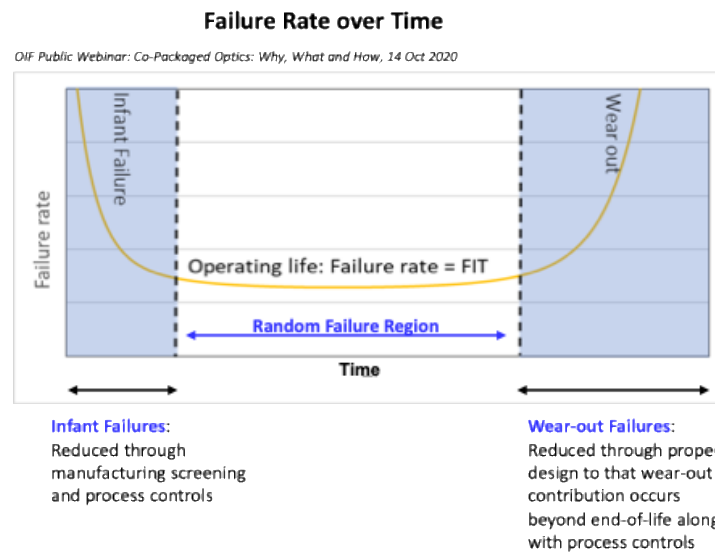


Figure 7-1 – Failure rate over time

Failures are often specified in units of “FIT” which represents the number of failures expected with 1e9 device-hours.

It is assumed that most or all the infant failures can be eliminated through a combination of process controls and manufacturing screening, so these failures will be largely eliminated before installation into a system.

Wear-out failures can be minimized through design and qualification, ensuring that significant wear-out only occurs after the random failure rate becomes dominant.

Random failures occur unpredictably and without a pattern and are typically modeled as a constant failure rate over time.

Given the complexity of a compute Pod with many elements and given the consequence of a failure as described above, the random failure target should be lower than typical Ethernet applications. In addition, it might make sense for the reliability target for co-packaged and on-board solutions to be higher than pluggable solutions due to increased time and cost to repair of these behind-the-front-panel architectures that are harder to service. This requires rigorous material selection, lifetime qualification at both component level and system level.

Scale-up links use mostly copper today and copper reliability sets the base failure rate expectation. As discussed in §1.3, replacing copper links with the current class of pluggable transceivers would negatively impact the reliability of the Pod. Older generations of discrete component assembled pluggable modules have failure rates of perhaps 50 FIT at best, normally driven by mechanical interfaces and thermal cycling stress, but newer generation of transceivers with integrated photonics, as well as improved manufacturing processes, suggest significantly lower failure rates are achievable. For example, Telcordia's SR332 "Reliability Prediction Procedure for Electronic Equipment" [13] highly integrated solutions can achieve superior reliability vs their discretely assembled counterparts. This is in part due to using a well-controlled integrated circuit process with C_{pk} of greater than 1.0 (a ± 3 -sigma control). A highly integrated solution also reduces the number of mechanical interfaces where thermal and mechanical stress can initiate various failure modes. A co-packaged transceiver would need a reliability level commensurate with silicon integrated circuits.

A related issue is the introduction of new technologies into scale-up links. It is suggested that new technologies demonstrate a track record in more fault tolerant applications before deploying in scale-up applications, especially in co-packaged or on-board locations. The initial deployment into environments with more fault tolerance offer error recovery mechanisms and making them suitable proving grounds. After a sustained track record of field reliability such new technologies be considered for co-packaged or on-board deployment in scale-up systems.

7.2.2 Repair

Periodically during operation, the Pod saves snapshots of the computation. Once a failure is diagnosed and repaired, the Pod can resume from the last saved snapshot. During the repair cycle, computation is halted stranding the investment in the Pod.

Repair is an important consideration, and the ease of repair is dependent on the architecture used as shown in Figure 2-1. A failed pluggable transceiver can more easily be replaced given the easy access to the front panel.

An on-board transceiver failure may be more time consuming as it requires the card to be removed and replaced. Later, a repair facility can replace the on-board transceiver.

A co-packaged transceiver failure also requires the card replacement. If the transceiver is soldered within the ASIC package, the board is likely more difficult to repair.

7.2.3 Importance of diagnostics

The availability of robust diagnostics is also an important consideration. A key metric is the time to repair as the entire Pod is down during the repair cycle. Being able to quickly diagnose and determine what needs to be repaired is an important part of the repair cycle.

Diagnostics provide important clues in determining the cause of the failure. For example, diagnostics can be used to determine if the problem is in the electrical interfaces or in the optical interfaces and

which end of the segment is at fault. The diagnostics can highlight link elements responsible for poor performance as well as pinpointing failed elements.

Although an compute element and switch may include robust diagnostics at either end of the link, low power transceiver solutions today often provide limited diagnostics. These low power transceiver solutions may improve their diagnostic capabilities for link segments and subsegments by providing additional diagnostics such as SNR, BER, and linearity. These capabilities have been realized using a linear redriver integrated with a sub-sampled digital eye monitor. The link management software and interfaces can be adapted to make use of these low power diagnostics capabilities.

7.2.4 Resiliency

Redundancy can be an effective path to reducing the failure rate. For example, introducing redundant lanes and components or redefining performance targets can effectively address these challenges with minimal overhead.

When a component fails, the transceiver can switch to using a spare component. There may be a short down time during the detection and switch over event but at least no components need to be replaced.

Redundancy promises significant improvements in reliability. For example, assume two redundant components each with a failure rate probability of p . The chance of both components failing would p^2 , a much smaller values than a probability p without redundancy. However, it is important to consider if the two components were made in the same production process and lot. If so, the failure rate of the backup component may have significant correlation with the failure of the primary component. Such correlation can significantly reduce the advantages of redundancy. The improvement from redundancy depends on the degree of correlation and of the particular distribution of the failure mode involved.

7.2.5 The importance of operating temperature

Component FIT rates increase exponentially with junction temperature as shown in Equation 7-1. The term E_A is the activation energy related the specific failure mechanism, k is Boltzmann's constant, T_{REF} is the reference temperature in kelvin that the FIT_{REF} was specified at, and T is the operating temperature in kelvin.

$$\text{EQ 7-1: } FIT_T = FIT_{REF} \cdot \exp\left(\frac{E_A}{k} * \left(\frac{1}{T_{REF}} - \frac{1}{T}\right)\right)$$

Depending upon the choice of technologies employed, the failure rate may be significantly improved when the optical components are kept cool, such as below 55°C—though this comes with thermal and power trade-offs. Operating at a higher temperature can significantly increase the annual failure rate.

7.2.6 Lasers

Optical transceivers need a laser source for the light generation. Many best-in-class lasers have a field based FIT rate in the range of a few FIT at 40 °C. The reliability contribution of multiple lasers in a transceiver can contribute a significant portion of the random failure rate. Higher failures rate for optical transceivers located behind the front panel is not desirable.

Historically, the light source has been viewed as the component in an optical link most likely to fail. A recent presentation by Meta at OCP 2025 [14] concluded that failures of pluggable modules were dominated by non-laser failure modes. Essentially the failure of the other components such as PCBA and wire bonds dominated the pluggable module failures. Suppliers of both edge-emitting lasers and VCSELs have claimed random FIT rates below 1, based on field data. As the industry transitions to highly integrated

solutions, the laser failure rate will become a larger portion of the transceiver failures. Designers can mitigate laser reliability risks by considering the quantity of lasers, their operating temperature and bias, and the ease of replacing a failed device.

Placing light sources behind the front panel makes them harder to service in the event of a failure and brings them closer to heat sources. Situating lasers in a module plugged into the front panel, such as the OIF's ELSFP [15], can create a better thermal environment and simplify the replacement of any failures. Using a thermoelectric cooler (TEC) in the module to limit the laser temperature can significantly improve the wear-out lifetime of the lasers, with a smaller improvement to the random failure rate, but comes at the expense of higher power consumption. A downside of the external pluggable approach is the increased optical coupling loss, which increases the optical power required from the lasers.

One approach to lower optical losses is to integrate lasers directly onto silicon photonics chips. Lasers fabricated after the heterogeneous bonding of III-V epi have been demonstrated to be reliable and require less optical power than external lasers. Though integrated lasers might be in a hotter operating environment than external ones, the fabrication of redundant lasers is straightforward. Adding redundant devices in cold standby can reduce the random FIT rate by orders of magnitude.

High-speed VCSELs are an energy-efficient option for scale-up links and have sparked interest for use in COI links. The high current densities required to achieve high bit rates tend to shorten the wear-out lifetimes at high temperatures, so some degree of cooling may be required for VCSEL solutions.

Aside from standard quantum well lasers, newer technologies such as quantum dot (QD) lasers and GaN micro-LEDs are being considered for AI applications. In theory, QD lasers could be beneficial in terms of power conversion efficiency and high temperature performance. QD lasers could also be advantageous in terms of reliability, as their strong carrier localization makes them less sensitive to crystalline defects. Interconnects based on micro-LED technology are attractive from an energy-efficiency perspective. The wide bandgap of GaN gives it intrinsic advantages in terms of reliability, but small emitters are more susceptible to defects created during fabrication and assembly.

High power lasers with acceptable FIT can be shared across multiple data ports to reduce cost/bit and FIT/bit contribution. For the next generation DWDM networks, laser sources require additional components like multiplexors. To continue support across multiple OEs with 16 wavelengths or more, further study on external laser formfactor may be needed.

7.2.7 Weight, Size, and Bend Radius: Serviceability & Safety Benefits

Compared to copper, optics offer significant advantages in weight, size, and bend radius. Current Cabled Backplane (CBP) designs can weigh up to 23 kg, requiring two-person lifts and raising the risk of a rack tip-over. With next-gen designs expected to double lane counts, CBP weight and size will increase further. Optics can help reduce overall CBP mass and footprint due to their lighter materials and more flexible routing, improving both serviceability and safety.

7.2.8 Qualification regimen

Historically, optical transceivers are qualified to the Telcordia standards including the well-known GR-468 as well as JEDEC standards. Applications which require higher reliability such as co-packaged silicon photonics may consider leveraging the Automotive Electronics Council qualification standards [16]. These standards leverage JEDEC and the MIL standards and set higher hurdles that is common for the optical industry. JEDEC is also developing reliability standards for silicon photonics solutions for telecom and datacom (part of JC 14.3).

8 Management

Ethernet pluggable transceivers widely use the OIF's CMIS for the management interface. CMIS has been extended to manage co-packaged systems where the laser source (an external laser) is physically separated from the optical transceiver.

While this approach is ideal for Ethernet switching solutions, scale-up optical links and especially co-packaged implementations using D2D interfaces may benefit from optimized approaches to transceiver management.

9 Summary

This white paper addresses the urgent need for energy-efficient, high-bandwidth, low-latency reliable optical interconnects in next-generation AI compute systems, which are increasingly composed of large, interconnected clusters of compute elements. Traditional copper-based interconnects are reaching their physical and energy limits, prompting a shift toward optical solutions. AI workloads require massive data movement across compute elements, with scale-up links utilizing an order of magnitude more bandwidth than scale-out links, interconnecting compute elements and switches within a Pod to support memory coherent communication while minimizing tail-latency.

The Compute Optics Interface white paper provides high level requirements and is intended to initiate further discussion on the many considerations and approaches for these scale-up optical links. It proposes a framework for energy-efficient photonic interconnects that balance performance, reliability, and serviceability and serves as a discussion foundation for OIF members to explore architectures, standards, and implementation strategies for future AI infrastructure.

A wide variety of technical avenues have been touched upon and there are clearly tradeoffs with each of the possible approaches. The best choice is dependent on the application's requirements.

One way to view the tradeoffs presented in this document are to consider several choices:

1. Required ASIC bandwidth edge escape density:
 - a. ≥ 2 T/mm: Consider CPO with a D2D electrical interface
 - b. ~ 1 T/mm: Consider CPC or CPO/NPO with a high-speed serial electrical interface
 - c. $< \sim 1$ T/mm: Consider off-package optical modules with a high-speed serial electrical interface
2. Reliability of the optical transceiver:
 - a. Consistent with SoP reliability, i.e. minor contribution to compute element or switch's multi-chip package reliability built with proven technologies: Consider CPO or on-board transceivers with innovative schemes (such as redundancy) to reduce the link interruption rate
 - b. Otherwise, consider pluggable transceivers with continued refinements with improved reliability
3. Power and latency requirements:
 - a. For best power and latency, consider non-retimed architectures while providing good link reliability, or retimed solutions with low-power, low-latency D2D interfaces.
 - b. For more challenging electrical channels with robust communication, consider Tx-retimed or fully retimed architectures

4. Optical modulation:

- a. Standardized optical link: Choose an optical standard, for example an IEEE specified link
- b. Non-standard optical link: Choose an optical approach which provides the required point-to-point link bandwidth and is consistent with the application's requirements

Of course, other factors not listed here such as overall solution cost would be important.

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Appendix A: Glossary

- 1) ASIC: Application specific integrated circuit
- 2) Bandwidth Edge Density: The aggregate full duplex bandwidth (Tx + Rx) divided by the edge width and expressed in units of Tb/s per mm.
- 3) BiDi, Bi-di, BiDir: Bidirectional optics with transmit and receive optical signals sharing the same fiber core
- 4) BoW: Bunch of wires is a D2D standard interface
- 5) BW: Bandwidth
- 6) CBP: Current cabled backplane
- 7) C2C: Chip to chip interface
- 8) CHI C2C: AMBA CHI Chip-to-Chip (C2C) Interface
- 9) Chiplet: A smaller chip co-packaged with other chips.
- 10) CMIS: Common management interface specification, [OIF-CMIS-05.3 – Common Management Interface Specification \(CMIS\) Revision 5.3](#) (September 2024).
- 11) COI: Compute Optics Interface used to interconnect local compute elements with optical links
- 12) CPC: Co-packaged copper which consists of a connector and cable
- 13) C_{pk} : Process capability index which measures how well a process is performing relative to the specification limits
- 14) CPO: Co-packaged optics
- 15) CRC: Cyclic Redundancy Check Is a low latency technique used to detect errors in the received data
- 16) CWDM: Coarse wavelength division multiplexed
- 17) CXL: Compute express link for efficient and coherent communication between CPUs and compute elements
- 18) D2D: Die to Die interfaces are typically short reach and often called “slow-wide” interfaces as opposed “fast-narrow” interfaces that utilize a high-speed SerDes
- 19) ELS: External laser source
- 20) ELSFP: External laser small form factor, an OIF interoperability agreement, <https://www.oiforum.com/wp-content/uploads/OIF-ELSFP-02.0-1.pdf>
- 21) Energy Utilization (EU): Energy utilization, often referred to as energy efficiency, is the power consumed divided by the unidirectional bandwidth excluding protocol overhead and expressed in units of pJ/b. Note that larger values indicate lower energy efficiency
- 22) Ethernet: A networking technology commonly used in data centers
- 23) FEC: Forward Error Correction is used to correct errors.
- 24) FIT: Failures in time typically specified as the number of failures in one billion hours of operation
- 25) Flit: Flow control unit is the smallest unit of data transported in a link.
- 26) GR-468: Refers to Telcordia’s general requirement for reliability
- 27) HBM: High bandwidth memory and is co-packaged with compute elements and CPUs.
- 28) ILD: Insertion Loss Deviation - the difference between the measured and fitted IL
- 29) IMDD: Intensity modulation direct detection
- 30) InfiniBand: A high-performance low latency interconnect used in data centers and AI compute
- 31) JEDEC: Joint electron device engineering council which produced various standards
- 32) Linear: An optical transceiver without a retimer (non-retimed) providing a one-to-one mapping between the electrical and optical signals

- 33) LPO: Linear pluggable optics is a front panel pluggable module that is linear
- 34) LRO: Linear receive optics is a front panel pluggable module that as a retimer in the Tx path but is linear in the Rx path. See RTLr for official OIF project name.
- 35) MLSE or MLSD: Maximum likelihood sequence estimation or detection
- 36) MZM: Mach-Zehnder modulator
- 37) NIC: Network interface card usually bridging the network connectivity with a CPU
- 38) NPO: Near packaged optics is an optical transceiver closely located to the ASIC, but not co-packaged with it
- 39) NRZ: Non-return to zero signaling sometimes referred to as PAM2
- 40) NVLink: A high-speed coherent link protocol for direct compute element to compute element, compute element to switch, or compute element to CPU connections and is proprietary to Nvidia.
- 41) O-Band: A wavelength range for data center optics centered around 1310 nm
- 42) OIF: Optical internetworking forum which publishes interoperability agreements for standardization
- 43) Optical Transceiver: Converts between electrical and optical signaling and may be co-packaged, on-board or in a pluggable form factor
- 44) PAM4: Four level signaling
- 45) PCIe: Peripheral component interconnect express is a standard defined by the PCI-SIG and used for high-speed connections between the computer's various components
- 46) PSM: Parallel single mode is an optical architecture with 1 λ per fiber
- 47) RS(544, 514): The FEC used in the Ethernet PHY that Ethernet, UEC, UALink, and SUE leverage
- 48) OSFP SerDes: Serializer / Deserializer
- 49) Pod: A collection of local compute elements and supporting hardware which is interconnected with links the provide for load/store memory semantics
- 50) RTRR: Retimed Transmit Retimed Receive is a fully retimed optical transceiver
- 51) RTLr: Retimed Transmit Linear Receiver is a half retimed optical transceiver with the retiming in the Tx path
- 52) Scale-up: Connectivity within a single system typically interconnecting compute elements and switches
- 53) Scale-out: Connectivity between a large number of single systems
- 54) SoP: System-on-Package
- 55) Slow-wide Interface: As opposed to a high-speed serial interface, it uses a large number or parallel lanes at slower speeds and is often singled ended
- 56) SR-332: Refers to Telcordia's special report for reliability prediction procedure for electronic equipment
- 57) SUE-T: Scale-up Ethernet is a proprietary scale-up link defined by Broadcom
- 58) TFLN: Thin film lithium niobate is a high-performance modulator material
- 59) TRO: Transmit retimed optics is half retimed and is also referred to as LRO or RTLr
- 60) UALink: Ultra accelerator link is an open standard for scale-up based on AMD's infinity fabric
- 61) UCle: Universal chiplet interconnect express is a standard D2D interface
- 62) UEC: Ultra Ethernet Consortium is defining a low latency interface for scale-up and scale-out
- 63) xGMI: Extended global memory interconnect by AMD is a high-speed coherent interconnect used to connect CPUs and compute elements